Causes and Effects of Employee Downsizing: A Review and Synthesis

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As employee downsizing has become increasingly ubiquitous in recent years, the study of this phenomenon has assumed greater significance. This article develops an integrative framework that incorporates environmental and organizational antecedents as well as the implications of downsizing for individuals and organizations. Key empirical studies are reviewed and major patterns and contradictions are identified. The authors identify and discuss theoretical and methodological concerns related to the extant literature and provide recommendations for future research aimed at developing a better understanding of employee downsizing.

Keywords: *employee downsizing; layoffs; redundancy; review*

The thing that people need to remember is that downsizing may be back on the front pages, but the downsizing never slowed down. Downsizing has been a constant and regular feature of the new working world, and it will continue to be.

—Bruce Tulgan, founder and CEO of RainmakerThinking

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Journal of Management, Vol. 36 No. 1, January 2010 281-348
DOI: 10.1177/0149206309346735
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Over the past couple of decades, employee downsizing has become an integral part of organizational life. Global competitive pressures coupled with ever-changing demand conditions have caused firms to critically examine their cost structures, including those associated with human resources. Once confined primarily to the United States, employee downsizing has, over time, become the norm in many countries. The magnitude of the downsizing activity has been exacerbated by the current recession. The extent of job losses has been staggering. In the United States alone, more than 6.5 million jobs have been "downsized" since the recession began in December 2007, with the numbers expected to grow in the foreseeable future (Krugman, 2009). In 2008, the U.S. Bureau of Labor Statistics recorded 21,137 "mass layoff events" in the United States involving more than 2 million job losses (Bureau of Labor Statistics, 2009).[1] Other countries have witnessed their share of job reductions, including countries that have traditionally shunned the practice of employee downsizing. The past year has witnessed significant employee reductions in countries such as Japan, Hong Kong, South Korea, and Taiwan. Even the economic juggernaut of China has been affected, with major layoffs in a number of sectors (e.g., among export-oriented firms and labor-intensive small businesses in China's coastal areas). Large-scale layoffs have also become commonplace in Europe, resulting in a spike in unemployment figures in many countries. For example, in Ireland, the unemployment rate jumped from 5.6% to 10.6% between March 2008 and March 2009, whereas Spain saw the figures go up from 9.5% to 17.4% over the same period. Companies in a broad range of industries (e.g., Continental AG, Luxotica, and Volvo) have resorted to massive employee downsizing over the past few months. Indeed, given its magnitude and impact, employee downsizing can legitimately be viewed as one of the most far-reaching and significant management issues of the current era.

Given the high incidence of employee downsizing and its implications from an economic and social standpoint, it is not surprising that the topic of employee downsizing has figured prominently in both the academic and popular business press. In addition to a proliferation of academic research, downsizing has also received extensive media coverage. More recently, "pundits" have questioned whether employee downsizing produces intended benefits in the form of lower costs and improved productivity. This concern has been voiced in a recent *Business Week* article by Vermeulen (2009) titled "The Tricky Truth About Downsizing." Vermeulen cites recent research by Guthrie and Datta (2008) to highlight the fact that although firms often view downsizing as a quick way of improving performance, the truth is that downsizing efforts often fail to achieve desired economic objectives.

Although downsizing can involve reductions in various combinations of physical and human resources or capital, our review focuses on *employee* downsizing, entailing personnel reductions in the organizational context (Cascio, 1993). More formally, after reviewing prior definitions (e.g., Cameron, 1994; Cameron, Freeman, & Mishra, 1993; Cascio, 1993; DeWitt, 1998; Freeman & Cameron, 1993; Kozlowski, Chao, Smith, & Hedlund, 1993), we define *employee downsizing* as follows: *Employee downsizing is a planned set of organizational policies and practices aimed at workforce reduction with the goal of improving firm performance.* Thus, we view downsizing as an intentional event involving a range of organizational policies and actions undertaken to improve firm performance through a reduction in employees.

Cameron (1994: 183) described downsizing as the "most pervasive yet understudied phenomenon in the business world." This viewpoint was later echoed by Budros (1999: 69), who

commented that "although downsizing has become an integral part of organizational life in the U.S. there is little serious theoretical or empirical work on this issue." Recognizing this void, researchers have generated a significant body of scholarly work on downsizing. Given the multifaceted nature of the topic, employee downsizing has been addressed by scholars in different disciplines (e.g., economics, finance, strategy, organization behavior, organization theory, human resources [HR], and sociology), using multiple theoretical and methodological perspectives. This article represents a "progress report" on the state of downsizing research. Our goal is to review, critique, and synthesize the relatively fragmented body of scholarship (involving 91 studies) that has emerged over the past 25 years (1984-2008) to provide a coherent and integrated portrait of extant research.

A synthesis and integration of this research is important and timely for several reasons. First, a systematic review of the literature based on an integrative framework can help identify areas of consensus and explore possible reasons for equivocal findings where consensus is lacking. Second, a review can pinpoint critical gaps in extant research. Doing so should facilitate and guide future research toward areas most in need of scholarly attention. Third, a systematic integration can help identify some context-specific generalizations that should prove useful for practicing managers. Fourth, a synthesis of existing research should help guide research on what has become an important topic in academic research. We have seen significant growth in employee downsizing research (as evident from the fact that 90 of the 91 studies in our 25-year review were published during the period 1990-2008). The dramatic increase in downsizing research in the United States in the 1990s can probably be attributed to the restructuring wave of the late 1980s. The oil shocks of the 1970s, along with the increased global (notably, Japanese) competition, forced many U.S. firms to reassess their cost structures, restructure their operations, and explore ways of reducing costs to regain competitiveness (Baumol, Blinder, & Wolff, 2003). The surge in the 1990s was also brought about by a changed managerial mind-set, as downsizing shed the stigma of being associated with organizational decline and began to be viewed by firms as a legitimate restructuring strategy aimed at improving organizational efficiency. With the current economic turmoil and the resultant upsurge in layoffs, the employee downsizing phenomenon is certain to attract significant scholarly attention in the near future.

For all the reasons discussed above, a review of the downsizing research at this time seems propitious. To keep the effort manageable, we limit our review to studies of employee downsizing taking place at the organizational level.[2] As such, studies that have examined downsizing at the economy or the industry level were excluded from the purview of this article. Instead, we focus on the *antecedents* to and the *consequences* of downsizing for individuals and organizations.

Theoretical Perspectives and Empirical Evidence

Much of the research on employee downsizing that has emerged over the past 25 years has centered on two major research questions: (a) What precipitates employee downsizing? and (b) What are the effects of downsizing from the standpoint of individual attitudes/behavior and organizational performance? In addressing these questions, scholars have invoked

multiple theoretical perspectives drawn from several disciplines. The phenomenon of downsizing can be viewed in the context of three sets of factors influencing organizational change: content, context, and process (Armenakis & Bedian, 1999; Self, Armenakis, & Schraeder, 2007). Although content is represented by "employee downsizing," it has several underlying characteristics, including (a) whether reductions are permanent or temporary (Hallock, 1998), (b) the adoption rate of employee reductions (Budros, 2004), (c) the level and scope of such downsizing (Iverson & Zatzick, 2007), and (d) the speed with which downsizing is executed (Budros, 2004). Although downsizing decisions are influenced by both external (environmental) and internal (organizational) contexts, downsizing outcomes (organizational and individual) are affected by factors related to the decision and implementation processes.

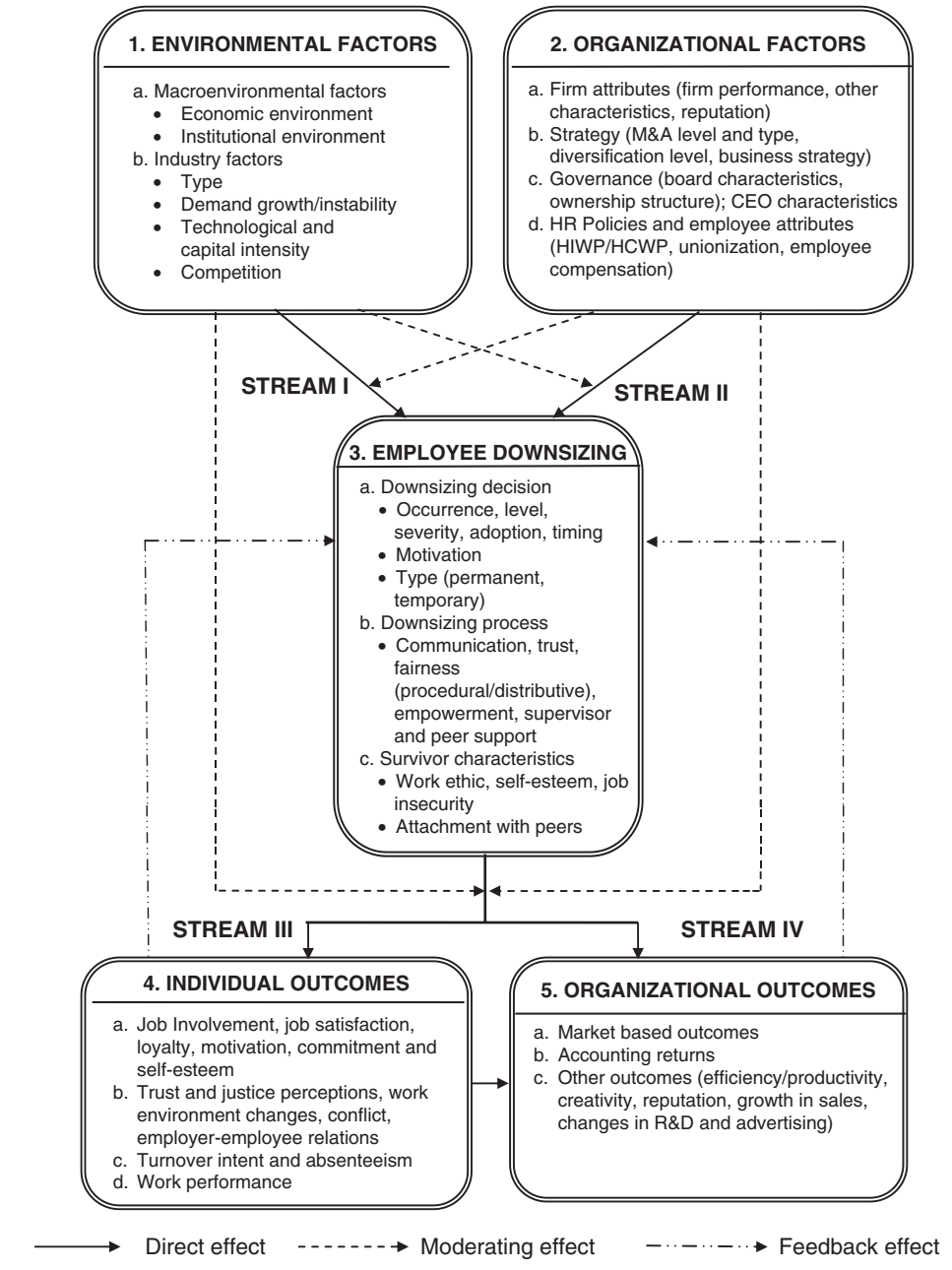
In this section, we review the theoretical and the empirical research on employee downsizing. We do so in the context of an integrative framework (Figure 1) that depicts the relationships between employee downsizing and its antecedents and consequences. The studies reviewed here were identified based on a systematic search of published research over the past 25 years (1983-2008) that have examined at least one of the linkages identified in Figure 1. The initial set of studies was identified using Google Scholar and the EBSCO Business Source Complete database. A careful examination of each study and references therein enabled us to identify additional studies. However, we limited this review to studies published in English. In other words, unpublished studies (e.g., working papers) and studies in other languages were excluded from this review.[3] Each study was then classified based on the framework of Figure 1. The process resulted in four broad streams of research. Streams I and II pertain to research on the environmental and organizational antecedents of employee downsizing, whereas Streams III and IV relate to research on the individual and organizational consequences of such downsizing. In the following paragraphs, we review the research that belongs to each of these streams.

Stream I: The Influence of Environmental Factors on Employee Downsizing

Much of this body of research is grounded in the industrial organization (IO) and institutional theory literatures. Employee downsizing has been linked to demand declines, with firms seeking to reduce labor costs while increasing fixed factor (capital) utilization (Cameron, Freeman, & Mishra, 1993). Demand declines often invoke market-clearing mechanisms that weaken and weed out underperforming firms, thereby causing significant employee layoffs. In addition, employee downsizing has been linked to structural changes brought about by technological advances, with increased productivity from such advances manifesting itself in the form of redundancies and employee layoffs. Moreover, industry deregulation and globalization have prompted many firms to seek efficiency and productivity enhancements via employee reductions in an effort to remain competitive (Budros, 1999).

Some theorists have viewed downsizing as an organizational response to institutional forces (e.g., Ahmadjian & Robbins, 2005; Budros, 1999; Freeman & Cameron, 1993; McKinley, Zhao, & Rust, 2000). Institutional theory posits that organizations are subject to pressures within an institutional field and that acquiring legitimacy requires firms to conform to institutional norms via the adoption of prevalent managerial practices (DiMaggio &

Figure 1
An Integrative Model of Employee Downsizing



Powell, 1983). According to this perspective, employee downsizing is seen as an outcome of the institutionalization process, whereby organizations seek legitimacy by mimicking downsizing practices by those firms viewed as being "successful" and "legitimate." Alternatively, arguments have been advanced that downsizing is the result of deinstitutionalization of existing practices (e.g., the employment-for-life practice that had been prevalent in Japan). However, abandonment of an institutionalized practice in favor of a new practice comes at the cost of diminished legitimacy, and such costs must be weighed against the benefits of abandonment. From an institutional theory standpoint, organizations may find safety in the number of firms adopting the new practice, allowing them to abandon the existing practice without incurring high costs (Ahmadjian & Robinson, 2001).

Two sets of studies belong to this stream of research. One set has examined the effects of industry characteristics (demand growth, industry structure) and changes therein on the decision to downsize. The other has used institutional theory arguments to assess how institutional pressures—regulatory, cognitive, and normative—influence downsizing. Our review identified a total of 12 studies belonging to Stream I, with 5 pertaining to downsizing in the United States and 7 related to downsizing in other countries. Studies belonging to this stream have been briefly summarized in Table 1 and are discussed in the following paragraphs.

Link 1a-3: Relationships between environmental conditions and downsizing. Although there is limited literature on how environmental conditions influence employee downsizing, some key findings emerge. For one, studies generally support the notion that employee downsizing is likely to be more prevalent under conditions of declining demand (Baumol, Blinder, & Wolff, 2003; Filatotchev, Buck, & Zhukov, 2000; Wagar, 1997). However, this view is not universal, with studies by Budros (1997, 2000, 2002) not finding any significant relationships between economic conditions and employee downsizing; indeed, he found that firms in the utility and industrial sectors often engaged in employee downsizing during economic peaks (Budros, 2000). Findings belonging to this stream also indicate that industry deregulation and privatization often trigger employee downsizing (Budros, 1997, 2002; Redman & Keithley, 1998). This is particularly true in the context of manufacturing firms. In addition, studies grounded in the institutional theory perspective generally indicate that firms are more likely to engage in significant employee reductions when there are increased levels of downsizing among referent firms in the population (Ahmadjian & Robinson, 2001; Budros, 1997, 2000, 2004; Tsai, Wu, Wang, & Huang, 2006). Furthermore, Budros (2000) notes that the institutional field influences downsizing, with mimicry playing a key role in the adoption of downsizing by firms. However, in a subsequent study, Budros (2002) found that this phenomenon exists among firms in the financial and utility sectors but not in the industrial (manufacturing) sector.

Link 1b-3: Relationships between industry characteristics and downsizing. Other studies belonging to this stream suggest that the proclivity to downsize and the level of downsizing are influenced by industry conditions. Coucke, Pennings, and Sleuwaegen (2007), for example, in their study of employee layoffs in Belgian firms, found that manufacturing firms were more likely to downsize than those in the service industry. Their findings were consistent with Budros' (1997, 2002) observation that downsizing was more prevalent among

(Text continues on p. 291)

Table 1
Stream I: Environmental Antecedents of Employee Downsizing

Study and Link	Sample	Variables	Findings
Budros (1997); Link: 1a-3a	Fortune 100 firms that downsized their labor force (1979-1994) identified from the *Wall Street Journal* (*WSJ*). Discrete-time event-history model using data from the Bureau of the Census.	Outcome: Downsizing adoption rate Explanatory: Industry deregulation (1 = *year deregulated*, 0 = *otherwise*), economic peaks/troughs (1 = *peak or trough*, 0 = *otherwise*), industry downsizing adoption (cumulative % of downsized firms between 1979 and 1994), industry culture (1 = *industrial and retail firms or competition-oriented*, 0 = *banking, diversified financial, insurance, and utility firms or service-oriented*)	Industry deregulation, economic peaks, industry downsizing adoption, and industry culture (industrial and retail sector) were positively associated with downsizing adoption rate.
Wagar (1997); Link: 1b-3a	1,140 medium- and large-sized Canadian firms that engaged in workforce reduction (1990-1992). Multinomial logistic regression analysis of survey data.	Outcome: Level of workforce reduction (mild: < 10% and layoffs less than one third of the time; moderate: < 10% and using layoffs more than one third of the time; severe: at least 10%) Explanatory: Market demand (6-point scale), industry type (retail, communication/transportation, finance and insurance, health care, education, and other business services)	Market demand was negatively associated with mild, moderate, and severe workforce reductions. Downsizing levels depended on the industry, with firms in communications undertaking mild reductions and avoiding severe reductions, health and education firms favoring mild to moderate reductions, and those in finance choosing mild reductions. Retail and other service firms did not show any significant propensity to implement workforce reductions.
Redman & Keithley (1998); Link: 1a-3a	4 Polish companies (1990-1996). Qualitative study based on interview data from the CEO and the senior personnel manager.	Outcome: Downsizing (count, %) Explanatory: Changes in the legal environment (enactment of the Employment Act in 1990)	The Employment Act of 1990, which increased privatization, deregulated wages and prices, and reduced state subsidies, also resulted in increased downsizing.

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Table 1 (continued)

Study and Link	Sample	Variables	Findings
Budros (2000); Links: 1a-3a, 1a-3a (1b)	Fortune 100 firms that downsized their labor force (1979-1995) identified from *WSJ*. Discrete-time event-history model using data from the Bureau of the Census and Standard & Poors (S&P) industry surveys.	Outcome: Downsizing adoption rate in utility, financial, and industrial sectors Explanatory: Economic peaks/troughs (1 = *peak and troughs*, 0 = *otherwise*), firm-based mimicry (cumulative % of Fortune 100 firms downsizing), and press-based mimicry (cumulative number of *Harvard Business Review* articles) Moderator: Industry type (utility, financial, and industrial)	Economic troughs did not affect downsizing for all types of firms. Although firms in the utility and industrial sectors downsized during economic peaks, financial firms did not. Although financial and utility firms downsized in response to firm-based and press-based mimicry pressures, industrial firms did not.
Filatotchev, Buck, & Zhukov (2000); Link: 1b-3a	103 privatized firms in Russia, Ukraine, and Belarus surveyed in 1994, 1996, and 1997. MANOVA of cross-sectional data from International Monetary Fund (IMF) country reports.	Outcome: Downsizing (% employee reduction) Explanatory: Industry decline (industry real output indexed as percentage of 1993 level), country of origin (Russia, Ukraine, or Belarus)	Declining industry conditions were positively associated with the extent of downsizing. Country of origin had no significant influence on downsizing.
Ahmadjian & Robinson (2001); Links: 1a-3a, 1a-3a (2a)	1,638 downsizing, publicly listed Japanese firms (1990-1997) identified from *Nihon Keizai Shimbun*. Panel data study using probit analysis of data from annual reports and *Recruit* magazine.	Outcome: Downsizing event (5% and 10% reduction of labor force) Explanatory: Downsizing in population (number of firms downsizing within the sample less focal firm over the past 3 years) Moderator: Firm size (log total assets), firm age (years since founding), firm reputation (times listed in *Recruit*'s 50 most attractive firms in 1980s), wage deviation from industry mean	Among firms downsizing at the 5% and 10% levels, higher downsizing levels by peer firms significantly increased the likelihood of downsizing. For those downsizing at the 5% level, the propensity to do so in response to layoffs by other firms was greater among larger, older, and more reputed firms. Among those downsizing at the 10% level, the propensity was higher among more reputable firms.

Budros (2002); Links: 1a-3a; 1a-3a (3b)	Fortune 100 firms that downsized their labor force (1979-1995) identified from *WSJ*. Discrete-time event-history model using data from the Bureau of the Census.	Outcome: Decrease in firm employment brought about by involuntary downsizing (1 = *decrease*, 0 = *otherwise*) Explanatory: Industry deregulation (1 = *year deregulated*, 0 = *otherwise*), economic downturn (reverse-coded annual real GNP), mimicry (% of Fortune 100 firms that implemented involuntary downsizing) Moderator: Industry type (1 = *manufacturing*, 0 = *otherwise*)	Industry deregulation was more likely to result in employee downsizing among manufacturing industry firms. Economic downturns and mimicry were not significantly related to employee downsizing.
Baumol, Blinder, & Wolff (2003); Link: 1b-3a	267 establishments in 20 manufacturing industries (1967-1997) from the Census of Manufacturing. Panel data study of secondary data from National Science Foundation, U.S. Department of Commerce, Bureau of Economic Analysis (BEA).	Outcome: Negative change in establishment employment Explanatory: Technical progress (total factor productivity growth), R&D intensity (ratio of the number of scientists and engineers engaged in R&D to total employment), degree of computerization (growth in office, computer, and accounting equipment per worker), foreign competition (period change in ratio of imports to gross output and exports to gross output), industry employment (sum of full-time employee equivalent in 2-digit manufacturing industry)	Change in industry employment was positively associated with changes in establishment employment. Among technical progress factors, only R&D intensity had a negative influence on establishment employment change. While foreign competition had a negative impact on establishment employment, there were no significant effects associated with the degree of computerization.

(continued)

Table 1 (continued)

Study and Link	Sample	Variables	Findings
Budros (2004); Link: 1a-3a	Fortune 100 firms that downsized their labor force (1979-1995) identified from *WSJ*. Discrete-time event-history model using data from Bureau of the Census.	Outcome: Downsizing adoption rate in the early (1979-1982) and late (1983-1995) phases Explanatory: Mimicry (cumulative % of downsized firms in each 2-digit standard industry classification [SIC] category), industrial firm (1 = *manufacturing*, 0 = *otherwise*)	Mimicry was positively associated with both early and late adoption of downsizing. Industrial firms were more likely to downsize in the early phase and less likely to do so in the later phase.
Frazier (2005); Links: 1a-3a, 1b-3a	788 Chinese state-owned enterprises (SOEs) (1994-1999). Ordinary least squares (OLS) regression of survey data.	Outcome: Workforce reduction in traditional, collective, and corporatized SOEs (% change in employment between 1994 and 1999) Explanatory: Location, industry (mining, machinery, utilities, food, chemicals, metals, furniture/paper, and textile), regulatory agency (central government or county/municipal), Chinese Communist Party (CCP) influence	Traditional SOEs located in Shanxi province and collective SOEs in the Sichuan province were positively associated with reduction of workforce. Industry, regulatory agency and CCP influence were not significant determinants of workforce reduction in traditional or collective SOEs. None of explanatory factors were associated with workforce reduction in corporatized SOEs.
Tsai, Wu, Wang, & Huang (2006); Link: 1a-3a	18 Taiwanese companies in 2003. Qualitative study based on interviews with top managers (VPs, HR managers) and employees (union leaders).	Outcome: Downsizing (count, %) Explanatory: Institutional factors (instruction from superior organization, downsizing as a commonly recognized strategy, and learning from benchmarked companies)	Institutional factors (downsizing as an accepted strategy and mimicry of benchmarked companies) motivated downsizing.
Coucke, Pennings, & Sleuwaegen (2007); Link: 1b-3a	339 Belgian firms reporting layoffs (1992-1994) to the Federal Planning Bureau. Multinomial logistic regression using data from questionnaire survey and company reports.	Outcome: Collective layoff (1 = *restructuring through layoffs* and 0 = *restructuring through exit or relocation*) Explanatory: Industry (1 = *manufacturing*, 0 = *service*)	Firms in the manufacturing sector were more likely to downsize via employee layoffs than those in the service sector.

manufacturing firms than nonmanufacturing firms. Budros (2004), however, observed that the relationship depended on the timing of the downsizing event, with manufacturing firms downsizing in the early phase of the study period (1979-1983) and nonmanufacturing firms doing so in the later phase (1983-1995). Finally, in a study involving Canadian firms, Wagar (1997) found that the level of downsizing varied across different sectors of the service industry, with employee reduction in the education and health sectors being harsher than that in the communications, finance, and retail sectors. Surprisingly, research on the role of other industry factors (e.g., industry capital intensity, industry research and development [R&D], and advertising intensity, concentration, and rivalry) has been conspicuous by its absence. The study by Baumol et al. (2003) remains an exception. They found that establishments in high technological intensity industries were less likely to experience employee downsizing. Furthermore, they observed that employee downsizing was directly related to foreign competition.

In sum, the limited Stream I research has left many questions unanswered. This translates into significant opportunities for future research aimed at understanding the role of environmental factors in determining the incidence and level of employee downsizing. Possible research questions for future research are discussed in the final section of the article.

Stream II: Organizational Influences on Employee Downsizing

The second body of research on the determinants of employee downsizing focuses on factors related to the organizational context as drivers of such downsizing. A key premise underlying this body of research is that the primary objective of employee downsizing is more efficient utilization of human resources. The organizational efficiency perspective draws on the resource-based view of the firm to argue that employee downsizing, with attendant cost savings, is a viable strategy in redressing organizational performance declines. In the context of underperforming firms, employee downsizing is seen as an important signal communicating organizational intentions and efforts at bridging the gap between stakeholders' expectations and achieved performance. On the other hand, layoff announcements made in the absence of poor performance may be perceived as a proactive measure aimed at maintaining current competitive advantage. Alternatively, it may signal private information held by managers of anticipated declining demand or dwindling firm economic health (Foster, 2002). The organizational efficiency perspective has also been used to justify downsizing in the aftermath of mergers and acquisitions (M&As). When similar firms merge, consolidation of operations generates personnel redundancies, undercutting financial performance. In such a context, employee downsizing represents a vehicle by which the merged entity can eliminate slack and realize operational synergies (O'Shaughnessy & Flanagan, 1998).

Research on the organizational antecedents of employee downsizing has also drawn extensively on agency theory arguments (e.g., Filatotchev et al., 2000; Perry & Shivdasani, 2005). Agency theory posits that weak governance mechanisms lead to managers engaging in activities that further their own interests at the expense of shareholders (Eisenhardt, 1989; Fama & Jensen, 1983). From this perspective, managers have a disincentive to downsize because of the pecuniary and nonpecuniary benefits derived from running larger and more

complex organizations and because any gains from downsizing accrue primarily to shareholders (Stulz, 1990). Effective monitoring (e.g., independent boards and greater institutional ownership) can mitigate agency problems and contribute to greater willingness on the part of managers to seek efficiency enhancements via employee reductions. It has been argued that the proclivity to downsize under conditions of weak performance is likely to be greater when the board is dominated by independent outsiders (Perry & Shivdasani, 2005). Likewise, institutional shareholders and blockholders, given their significant ownership stakes and voting power, are likely to be more aggressive in monitoring management and requiring top management to employ downsizing options if they believe that doing so will enhance firm value. In addition to monitoring, agency theory arguments highlight the role of incentive mechanisms in addressing agency problems. It is based on the argument that when the interests of managers are closely aligned with those of shareholders (via equity ownership and/or a compensation structure), there will be a greater propensity on their part to seek efficiency enhancements from employee reductions.

The event of downsizing can also be attributed to changes in prevailing governance mechanisms. In countries (e.g., Japan and Korea) where a stakeholder orientation is being replaced by a shareholder orientation, the changes in cultural norms governing employment practices have contributed to increased levels of employee downsizing (Ahmadjian & Robbins, 2005; Vicente-Lorente & Suárez-González, 2007). Finally, the incidence and level of employee downsizing can also be explained in the context of the upper echelons perspective (Hambrick & Mason, 1984). Scholars using this framework (e.g., Hallock, 1998) have argued that managers' demographic characteristics (e.g., tenure, age, functional background) represent proxies for their cognitive orientation and knowledge base and, consequently, influence strategic decisions such as downsizing.

The 35 studies in Stream II can be broadly grouped into four categories: (a) those examining the effects of firm attributes (including firm performance) on employee downsizing (Link 2a-3 with 21 studies), (b) those seeking to assess the influence of firm strategy (e.g., M&As, firm diversification) on downsizing (Link 2b-3 with 11 studies), (c) those examining how governance structures (board characteristics, ownership, compensation structures, CEO characteristics) help determine the incidence and level of downsizing (Link 2c-3 with 15 studies), and (d) research on the implications of firms' HR policies for downsizing events (Link 2d-3 with 14 studies). Table 2 provides a summary of Stream II studies. Of the 35 studies belonging to this stream, 20 have been based on downsizing in the United States, with the remaining 15 related to downsizing events in other countries.

Link 2a-3: Relationships between firm attributes and downsizing. Based on the conventional view that employee downsizing reduces labor costs and boosts profitability, several studies belonging to this link have examined the effects of firm performance (accounting, market-based, and efficiency measures) on downsizing decisions. There is general agreement across studies that firm performance is an important factor in downsizing decisions, with performance declines often resulting in employee reductions (Ahmadjian & Robinson, 2001; Baumol et al., 2003; Budros, 2002, 2004; Coucke et al., 2007; Hillier, Marshall, McColgan, & Werema, 2006; Kang & Shivdasani, 1997; Tsai et al., 2006; Yu & Park, 2006). However, a few studies (Iqbal & Shetty, 1994; Perry & Shivdasani, 2005; Yoo & Mody,

(Text continues on p. 305)

Table 2
Stream II: Organizational Antecedents of Employee Downsizing

Study and Link	Sample	Variables	Findings
Bethel & Liebeskind (1993); Link: 2c-3a	93 randomly selected Fortune 500 firms that restructured (1981-1987). Ordinary least squares (OLS0 analysis of secondary data from *Value Line Investment Survey* and Compustat.	Outcome: Downsizing (% employee reduction) Explanatory: Insider ownership (% of total outstanding common voting shares owned by insiders), blockholder ownership (% of total shares held by owners of 5% or more common shares), institutional ownership (% of total shares owned by institutions)	Blockholder ownership was positively associated with employee downsizing. No significant relationships were observed between institutional and insider ownership and employee downsizing.
Ofek (1993); Link: 2a-3a	358 firms with layoffs that experienced a year of above-average returns followed by a year of poor performance (1983-1987) identified from *Wall Street Journal* (*WSJ*). Logistic regression using data from Lexis and Center for Research in Security Prices (CRSP) databases.	Outcome: Layoffs (10% or more employee reduction over the year) Explanatory: Leverage (book value of debt/ [book value of debt + market value of equity])	Highly leveraged firms that experienced short-term distress exhibited a greater propensity to engage in employee downsizing.
Mishra & Mishra (1994); Link: 2c-3a	91 business units of 43 North American automotive firms surveyed in 1990-1991. OLS analysis of data from top executives, managers, customers, suppliers.	Outcome: Workforce reduction Explanatory: Mutual trust (a) within the top management team (TMT), (b) between TMT and employees, and (c) between business units and their customers and suppliers	Mutual trust between the TMT and employees was negatively associated with workforce reduction, while mutual trust within the TMT and between business units and their customers and suppliers was positively related to workforce reduction.
Iqbal & Shetty (1994); Link: 2a-3a	75 firms announcing layoffs (1986-1989) identified from *WSJ*. Univariate *t* tests based on Compustat data.	Outcome: Layoff announcements Explanatory: Earnings performance (net income/sales), sales performance (change in sales), labor intensity (labor cost/sales), and stock price performance (change in	Current ratio and inventory levels were positively associated with layoff announcements. Earnings performance, sales performance, labor intensity, stock price performance, slack resources,

(continued)

Table 2 (continued)

Study and Link	Sample	Variables	Findings
		stock price), current ratio, slack resources (long-term debt/assets, earnings before interest and taxes/interest, current ratio, working capital/sales), level of inventory (finished goods/employee), change in annual dividends in year prior to layoff, average wage level (labor cost/employee) and investment in human capital (invested capital/employee)	change in dividends, wage level, and investment in human capital did not significantly influence layoff decisions.
Budros (1997); Links: 2a-3a, 2b-3a, 2c-3a, 2d-3a	Fortune 100 firms that downsized their labor force (1979-1994) identified from *WSJ*. Discrete-time event-history model using data from *M&A Journal*, *Fortune*, Hoover's, Inc., *SEC Bulletin*, Compact Disclosure, *Moody's Manuals*.	Outcome: Downsizing adoption rate Explanatory: Change in market share (firm's assets/industry assets), change in shareholder value (annual % change in stock price and dividend yield), firm productivity (net income/employee), new technology (annual % change in PP&E), firm size (number of employees), firm age, number of domestic and foreign mergers and acquisitions (M&As), takeover tries, ownership status (if publicly traded), debt ratio (firm debt equity ratio - mean debt equity ratio for all Fortune 100 firms in that industry), compensation/employee (logged wage, salary, and benefits/employee)	Firm size, decreasing market share, decreasing shareholder value, and ownership status were positively associated with downsizing adoption rate. Productivity was negatively related to downsizing adoption. Firm age or new technology did not affect downsizing adoption rate. While domestic M&As did not influence downsizing adoption, foreign M&As had a positive impact on downsizing adoption rate. Takeover bids, debt ratio, and average employee compensation had no significant effect on downsizing adoption rate.
Kang & Shivdasani (1997); Link: 2a-3a	92 Japanese firms that experienced significant performance declines (1986-1990) identified from various sources.	Outcome: Layoff likelihood Explanatory: Return on asset (ROA), firm size, firm leverage	The likelihood of significant employee layoffs was higher among larger firms and firms with lower ROA. Firm leverage was not associated with employee layoffs.

	Logistic regression of data from the Pacific-Basin Capital Markets Research Center database.		
Wagar (1997); Links: 2a-3a, 2b-3a, 2d-3a	1,140 medium- and large-sized Canadian firms that engaged in workforce reduction (1990-1992). Multinomial logistic regression analysis of survey data.	Outcome: Workforce reduction (mild: less than 10% and layoffs less than one third of the time; moderate: less than 10% and using layoffs more than one third of the time; severe: at least 10%) Explanatory: Merger (1 = *involvement in M&A*, 0 = *otherwise*), change in strategy (1 = *major change in strategy*, 0 = *otherwise*); firm size, unionization (1 = *union*, 0 = *no union*), training (work groups with training program/total number of work groups), commitment to job security, introduction of new technology (1 = *presence*, 0 = *absence*)	Merger involvement was not significantly associated with workforce reduction. Major changes in strategy were associated with severe workforce reductions but not with mild or moderate reductions. Firm size positively influenced mild and moderate workforce reductions but not severe reductions. Firm commitment to job security was negatively associated with moderate and severe reductions ($p < .01$) but not mild reductions. There were no significant relationships between unionization or training and workforce reductions. Introduction of labor-saving technology positively affected all levels of workforce reduction.
O'Shaughnessy & Flanagan (1998); Link: 2b-3a	136 large mergers and acquisitions (1989-1993) identified from *M&A Almanac*, LexisNexis, and *WSJ*. Logistic regression of cross-sectional data obtained from Securities Data Company (SDC), *M&A Magazine*, Compact Disclosure.	Outcome: Merger-induced layoff announcements Explanatory: Target firm performance (ROIC) and operating efficiency (revenue/employee) prior to the merger, merger relatedness, cross-border acquisitions (1 = *target and bidder firm listed different countries as headquarters*, 0 = *otherwise*), merger financing (1 = *financed with borrowing*, 0 = *otherwise*)	Target firm operating efficiency was negatively associated with postmerger layoffs. No relationship was observed between target firm performance and postmerger layoffs. Related acquisitions were more likely to be followed by layoffs. No significant relationships observed between cross-border acquisitions and layoffs. Debt financing did not significantly influence the likelihood of merger-induced layoffs.

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Table 2 (continued)

Study and Link	Sample	Variables	Findings
Hallock (1998); Links: 2a-3a, 2c-3a	550 U.S. firms with layoff announcements (1987-1995) identified from *WSJ*. Univariate *t* tests using secondary data from *Forbes* and Compustat.	Outcome: Layoff announcement Explanatory: CEO age, CEO firm tenure, tenure as CEO, CEO pay (total compensation), change in CEO pay in the year prior to the layoff, firm size (market value of equity, total number of employees)	CEO tenure was significantly lower in firms announcing layoffs than that in firms without layoff announcements. No significant differences in CEO age, CEO firm tenure, total CEO compensation, or percentage change in CEO pay were observed between layoff and nonlayoff firms. Firm size was positively associated with layoff announcement.
Budros (2000); Links: 2a-3a (1b), 2b-3a (1b), 2c-3a (1b), 2d-3a (1b)	Fortune 100 firms that downsized their labor force (1979-1995) identified from *WSJ*. Discrete-time event-history model using secondary data from *Fortune*, *M&A Journal*, *Moody's Manuals* and *Who's Who in Finance and Industry*.	Outcome: Downsizing adoption rate Explanatory: Market share decline, profitability, reductions in shareholder value, M&A activity (annual number of M&As by firm), new CEO (1 = *firm hired new CEO*, 0 = *otherwise*), CEO finance background (1 = *finance background*, 0 = *otherwise*), average employee compensation (logged measure of wages, salaries, and benefits per employee) Moderator: Industry type (utility, financial, and industrial)	Market share decline was positively associated with downsizing in all industries. Low profitability and declining shareholder values resulted in downsizing among financial firms but not in industrial and utility firms. M&As were positively associated with downsizing among industrial firms but not in the financial and utility sectors. CEOs with finance background were associated with the adoption of downsizing in financial firms but not in industrial and utility firms. The hiring of a new CEO increased the likelihood of adoption of downsizing in industrial firms but not in financial and utility firms. Employee compensation was not a significant factor in the adoption of downsizing at firms in any of the industry types.

Osterman (2000); Links: 2a-3a, 2d-3a	462 establishments in the U.S. that were in business in 1992 and 1997. Logistic regression of survey establishment data from National Survey of Establishments.	Outcome: Layoff (1 = *>5% reduction in work force in 1995 and 1996,* 0 = *otherwise*) Explanatory: High-performance work organization (HPWO) practices, firm size (number of employees), firm age, sales growth (%), skill level of core employees (1 = *highly skilled*, 0 = *otherwise*), unionization (1 = *some employees at establishment covered by collective bargaining*, 0 = *otherwise*), presence in international markets (1 = *competes in international markets*, 0 = *otherwise*)	HPWO practices were positively associated with layoffs. Firm age, sales growth, and skill level of core employees were negatively associated with the likelihood of layoffs. The likelihood of layoffs was also lower when the establishment competed in international markets. The relationship between unionization and layoffs was negative but not significant.
Filatotchev, Buck, & Zhukov (2000); Link: 2c-3a	103 privatized firms in Russia, Ukraine, and Belarus surveyed in 1994, 1996, and 1997. MANOVA of survey data from firm managers.	Outcome: Employment change Explanatory: Institutional ownership, percentage voting shares held by outside private individuals, managerial ownership	Outside ownership was positively associated with downsizing, while managerial ownership discouraged downsizing. Institutional ownership did not affect firm downsizing. Change in managerial ownership was positively related to downsizing.
Iverson & Pullman (2000); Links: 2a-3a, 2d-3a	415 staff members across a hospital in Victoria, Australia. Survey administered 2 weeks before the announcement of first lot of employee redundancies. 51 of the staff members were laid off and 170 resigned voluntarily. Event-history analysis using survey data.	Outcome: Layoffs Explanatory: Gender (1 = *female*, 0 = *male*); full vs. part time, blue collar (1 = *blue collar*, 0 = *white collar*); education (years), union affiliation (1 = *union member*, 0 = *otherwise*), absenteeism (days absent), job opportunity, promotional opportunity, job security, coworker and supervisory support, work	Age and full-time status were both positively associated with layoffs. Coworker support and work overload were both negatively associated with layoffs. Employees who felt that conditions in the hospital had deteriorated were more likely to be laid off. Other factors were not significantly associated with downsizing.

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Table 2 (continued)

Study and Link	Sample	Variables	Findings
		overload, role overload, intent to leave, reaction to amalgamation following merger (worse or better)	
Yoo and Mody (2000); Links: 2a-3a, 2d-3a	50 U.S. local telephone companies with 90% of the market (1988-1995). OLS regression of panel data from the Federal Communication Commission's (FCC) *Statistics of Communications Common Carriers*.	Outcome: Change in employment level from the previous year Explanatory: Productivity (log of ratio of number of calls/number of employees), profit rate (log of net operating revenues/ total value of telecom plant in service), firm size (log of total operating revenues), mean employee compensation (log of total compensation/total number of employees), new technology	Decline in productivity was positively associated with employment reductions in the short and the long term. Firm profitability and firm size were not related to employment reductions. Firms with higher mean employee compensation were more likely to reduce employment levels in the short term. New technology was also positively associated with downsizing in the short term.
Ahmadjian & Robinson (2001); Links: 2a-3a, 2c-3a, 2d-3a	1,638 publicly listed Japanese firms (1990-1997) that downsized identified from *Nihon Keizai Shimbun*. Probit analysis of panel data obtained from annual reports and *Recruit* magazine.	Outcome: Downsizing event (involving 5% and 10% of labor force) Explanatory: ROA, sales growth, losses (over previous 5 years), profits/employee, firm size (log of total assets), firm age, and firm reputation (times listed on *Recruit*'s 50 most attractive firms in 1980s), foreign ownership (% shares held by non-Japanese shareholders), wage deviation from industry mean	Firm age, foreign ownership, firm losses, and declining sales were all positively associated with downsizing at both the 5% and 10% level. Firm reputation was negatively associated with downsizing at the 10% level. Firm size and wage deviation from industry mean did not have any influence on firm downsizing at either level.
Budros (2002); Links: 2a-3a, 2b-3a, 2c-3a	Fortune 100 firms that downsized their labor force (1979-1995). Discrete-time event-history model using data from *Fortune*, *Forbes*, *M&A Journal*, and *Who's Who in Finance and Industry*.	Outcome: Decrease in employment via involuntary downsizing Explanatory: Profit decline (annual % decline in net profits or income), M&A activity (annual number of M&As by	Decline in profits and M&A activity contributed to employee downsizings. However, over time, the influence of profit decline and M&As on involuntary downsizings declined. Firms with younger

		firm), CEO background (1 = *operations background*, 0 = *otherwise*), CEO age, and CEO tenure	CEOs and CEOs with operations backgrounds were more likely to institute downsizing. CEO tenure was not related to downsizing.
Conyon, Girma, Thompson, & Wright (2002); Link: 2b-3a	442 mergers by 277 U.K.-based firms (1967-1996) identified from London Share Price Database and Cambridge/DTI Databank of Company Accounts. Univariate *t* tests (difference in means) using secondary data from Datastream.	Outcome: Percentage change in employment in the merged firm 1, 2, 3, and 4 years after the merger from the combined values of the employment of the two firms prior to the merger Explanatory: Type of M&A (related/unrelated; friendly/hostile)	Related and hostile M&As resulted in significant employment declines in all the four postmerger time periods. Friendly mergers experienced a significant decline in employment only in the first year following the merger. No significant decline in employment levels was observed for unrelated M&As.
Baumol, Blinder, & Wolff (2003); Links: 2a-3a, 2d-3a	267 establishments in 20 manufacturing industries (1967-1997). Panel data regression analysis of secondary data from National Income and Product Accounts, Bureau of Labor Statistics, Department of Commerce, and BEA.	Outcome: Downsizing (change in establishment employment) Explanatory: Change in profitability (lagged profit), growth in total capital/worker, unionization rate (ratio of employees covered by union contracts)	Declining profitability was positively associated with employee downsizing. Unionization had a negative influence on downsizing, and growth in total capital/worker was not related to employment reductions.
Budros (2004); Links: 2a-3a, 2b-3a, 2c-3a	Fortune 100 firms that downsized their labor force (1979-1995) identified from *WSJ*. Discrete-time event-history model using secondary data from annual reports, *Fortune*, *Forbes*, *M&A Journal* and *Who's Who in Finance and Industry*.	Outcome: Downsizing adoption rate in the early (1979-1982) and late (1983-1995) phases Explanatory: Profit decline (annual % decline in net profits or income), shareholder value decline (annual change in firm's stock price and dividend yield), level of mergers and acquisitions activity (annual number of M&As), CEO background (1 = *finance or accounting*, 0 = *otherwise*), CEO age, CEO firm tenure	Profit declines positively influenced early adoption of downsizing, while declines in shareholder values positively influenced late downsizing adoption. Level of M&A activity did not influence the adoption of downsizing in either phase. Firms with younger CEOs and CEOs with longer tenures were associated with early adoption of downsizing, while finance and accounting background was associated with late adoption of downsizing.

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Study and Link	Sample	Variables	Findings
Ahmadjian & Robbins (2005); Link: 2c-3a	1,108 nonfinancial, publicly listed Japanese firms (1991-2000). Probit analysis of panel data from the Nikkei NEEDS database.	Outcome: Downsizing event (involving 5% and 10% decreases in labor force) Explanatory: Foreign ownership (percentage of shares held by non-Japanese shareholders), financial ownership (percentage shares held by Japanese banks, trust banks, and insurance companies), group membership (in Sumitomo, Mitsubishi, Mitsui, Fuyo, Sanwa, or DKB groups)	At the 5% level, group membership had a positive influence on downsizing. No significant relationships were present between foreign ownership or financial ownership and downsizing. However, group and financial ownership interacted with foreign ownership to reduce the likelihood of downsizing. At the 10% level, none of the ownership types were significant predictors of firm downsizing.
Perry & Shivdasani (2005); Link: 2a-3a	94 nonfinancial firms in the United States that downsized in 1993 after financial decline identified from *WSJ*. Logistic regression based on data from Compustat.	Outcome: Employee reduction Explanatory: Firm size, percentage change in ROA, market-to-book ratio, leverage, board size, % equity ownership of independent directors, % equity ownership of officers and affiliated directors, outside board indicator (>50% of board are independent outside directors).	Firm size, change in ROA, and leverage were not significantly associated with employee reductions. Market-to-book ratio was negatively associated with employment reductions. Firms with outside boards were more likely to engage in employee reductions. Board size and equity ownership of independent directors and insiders were not significant predictors of employment reduction.
Frazier (2005); Links: 2a-3a, 2d-3a	788 Chinese state owned enterprises (SOEs) (1994-1999). OLS regression of survey data.	Outcome: Workforce reduction in traditional, collective, and corporatized SOEs between 1994 and 1999 Explanatory: Firm output value, tax revenue (log amount remitted by sample SOEs), employee composition (% temporary, % women, and % retirees), mean wage, hiring autonomy (single item instrument)	There was no relationship between firm output and downsizing in traditional and collective SOEs. Reduction of workforce in traditional SOEs was greater when tax revenues were higher. In traditional and collective SOEs, a lower proportion of retirees resulted in greater likelihood of workforce reduction. Mean wage and hiring

			autonomy, along with the percentage of temporary and women workers, did not influence workforce reductions.
Gittell, Cameron, Lim, & Rivas (2006); Links: 2a-3a, 2d-3a, 2d-2a-3	10 major U.S. airlines (1987-2005) that downsized as identified in press announcements. OLS regression of data from Yahoo Finance, Thomson Financial, CNNfn, the National Mediation Board, and USDOT.	Outcome: Layoffs in response to 9/11 crisis Explanatory: Debt/equity ratio, days of cash on hand, number of strikes and releases Mediator: Efficiency, employee productivity, aircraft productivity (block hours per aircraft day)	No relationship observed between cash on hand and layoffs. Debt ratio and unit cost per seat mile were positively associated with layoffs. Employee and aircraft productivity mediated the relationship between strikes and releases and layoffs, with strikes and releases reducing both employee and aircraft productivity and, thereby, contributing to increased layoffs.
Yu & Park (2006); Link: 2a-3a	258 publicly traded firms in Korea (1997-1999). OLS regression of survey data from the Korea Investors Service-Financial Analysis System (KIS-FAS) database.	Outcome: Downsizing (1 = *firm implemented downsizing or honorable retirement or both*, 0 = *otherwise*) Explanatory: ROA, asset turnover, operating income (OI)/employee, firm sales, sales/employee, value added/employee (value added = OI + labor costs + depreciation + rent + taxes and dues)	Lower ROA, operating income/employee, and asset turnover were all associated with downsizing in 1998 and 1999. Firm sales, sales per employee, and value added per employee were not significantly associated with downsizing.
Hillier, Marshall, McColgan, & Werema (2006); Links: 2a-3, 2b-3, 2c-3, 2d-3	322 layoff announcements by firms listed in the London Stock Exchange (LSE) (1990-2000) identified from FT Extel Company Research and Lexis-Nexis databases. Logistic regression using data from Datastream.	Outcome: Layoff announcements (involving >0.1% of labor force) Explanatory: Firm market value (log of equity at financial year-end prior to layoff), firm diversification (Herfindahl Index on 3-digit SIC lines), threats to managerial control (takeover threat and	Market value and firm diversification were positively associated with the likelihood of firm layoffs. Significant declines in firm profitability and increases in debt increased the likelihood of employee layoffs. Threats to corporate control were also positively associated with layoffs.

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Study and Link	Sample	Variables	Findings
		CEO turnover), change in ROA, employee productivity (sales/employee), and leverage (change in debt/asset ratio and interest coverage)	Drop in employee productivity did not result in layoff announcements.
Yawson (2006); Link: 2c-3a	1,501 firm-year observations from U.K. firms listed on the LSE (1992-2003). Logistic regression using data from Datastream.	Outcome: Layoff event (1 = *firms with at least 20% reduction in workforce in 2 consecutive years*, 0 = *otherwise*) Explanatory: Board size, proportion of outside directors (ratio of outside directors to total number of directors), director compensation	Compared to nonlayoff firms, firms that experienced layoffs had smaller boards, higher proportion of outside directors, and higher director compensation.
Tsai, Wu, Wang, & Huang (2006); Links: 2a-3a, 2c-3a	18 Taiwanese companies in 2003. Qualitative study based on interviews with top managers (VPs, HR managers) and employees (union leaders).	Outcome: Downsizing (count, %) Explanatory: Change in profitability (stagnant profitability or decline), social cognition processes	Declining or stagnant profitability was positively related to employee downsizing in sample firms. Social cognition factors such as the actualization of manager's knowledge and espousal of reengineering processes influenced the motivation to downsize.
Krishnan, Hitt, & Park (2007); Link: 2b-3a	174 layoff firms involved in related acquisitions (1992-1998) identified in *WSJ*. OLS regression using CRSP data.	Outcome: Workforce reduction in merged firms (% reduction in number of employees in combined organization) Explanatory: Acquisition premium (% difference between final purchase price and trading price of target's stock)	Acquisition premiums were positively associated with postmerger workforce reductions in related acquisitions.
Brookman, Chang, & Rennie (2007); Link: 2c-3a	Layoffs announcements by 484 firms (1993-2003) in *WSJ*. Conditional logit regression using data from Execucomp, proxy statements, and CRSP.	Outcome: Layoff announcements Explanatory: CEO equity portfolio incentives (change in CEO wealth from $1 change in stock price), CEO equity ownership	There was a positive relationship between CEO equity portfolio incentives and layoffs. Equity ownership, however, was not related to layoff announcements.

Coucke, Pennings, & Sleuwaegen (2007); Links: 2a-3a, 2b-3a	339 Belgian firms reporting layoffs (1992-1994) to the Federal Planning Bureau. Multinomial logit regression based on data from surveys and company reports.	Outcome: Collective layoff (1 = *restructuring through layoff*, 0 = *restructuring through exit or relocation*) Explanatory: Capital/employee, scale economies (minimum efficient scale), return on equity (ROE), firm size (log of firm's added value), firm age, interaction of leverage and financial distress (1 = *ROE* < -0.08; 0 = *otherwise*), multinational production network (1 = *multinational enterprise (MNE) with multinational production network*, 0 = *otherwise*), recent investments in tangible assets, product differentiation	Firms with lower ROE and high scale economies were more likely to engage in employee layoffs. Larger and younger firms were also more likely to restructure via employee layoffs. Higher leverage in financially distressed firms was positively associated with employee layoffs while higher product differentiation and recent investments in tangible assets reduced the probability of such layoffs. Capital per employee was not a determinant of such layoffs and the presence of a multinational production network also did not influence layoff decisions.
Stavrou, Kassinis, & Filotheou (2007); Links: 2c-3a, 2c-3a (2d)	Layoffs in 102 family firms (2000-2002) based on announcements in *New York Times* and *WSJ*. Hierarchical regression of data from EDGAR, Compustat, and KLD databases.	Outcome: Layoff ratio (number of downsized employees/total workforce) Explanatory: Ownership (1 = *if at least 2 directors have a family relationship and family members own or control at least 5% of voting stock*, 0 = *otherwise*) Moderator: Firm employee- and community-friendly policies (based on six indicators), ROA	Family businesses engaged in lower levels of downsizing compared to non–family businesses. Financial performance and employee- or community-friendly practices did not moderate the relationship between ownership and level of downsizing.
Iverson & Zatzick (2007); Link: 2d-3	1,939 workplaces in Australia with more than 20 employees (1995-1996). Ordinal logistic regression using data from the Australian Workplace Industrial Relations Survey.	Outcome: Downsizing harshness (5 levels ranging from no layoffs to compulsory layoffs) Explanatory: High commitment work practice (HCWP) index	Firms with a higher HCWP index were more likely to downsize but did so in a less harsh manner and avoided the use of compulsory layoffs.

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Table 2 (continued)

Study and Link	Sample	Variables	Findings
Vicente-Lorente & Suárez-González (2007); Link: 2c-3a	425 largest Spanish firms (1990-1998). Panel data study using probit model of secondary data from *Fomento de la Produccíon*.	Outcome: Downsizing event (>5% decrease; 1 = *if firm has downsized in a given year*, 0 = *if it had not*) Explanatory: Ownership (foreign nonlisted, domestic listed, state-owned nonlisted, foreign listed, state owned listed, state-owned to be privatized)	Domestic listed, foreign listed, state-owned listed, and state-owned to be privatized ownerships were all positively associated with downsizing. There were no significant relationships between foreign nonlisted and state-owned nonlisted ownership and downsizing.
Lehto & Böckerman (2008); Links: 2b-3a, 2b-3a (1b)	7,923 M&As involving Finnish firms (1989-2003) identified in *Talouselämä*. Panel data study using probit model of secondary data from the National Board of Patents and Registration of Finland and *Business Register* by Statistics Finland.	Outcome: Reductions in employment in the short term (between the merger year and the preceding year) and long term (between the premerger year and 2 years after the merger) Explanatory: Cross-border M&As, domestic M&A with domestic acquirer, domestic M&A with foreign acquirer, internal restructuring (management buyouts) Moderator: Industry–manufacturing (including utility), construction and other services, and trade (hotels and restaurants)	Employee reductions resulted from cross-border M&As and domestic M&As with domestic acquirers. Employment reductions involving internal restructurings was positive and significant in the short term but not in the long term. Construction industry firms experienced employment declines in domestic M&As with domestic acquirers. Declines in employment due to cross-border M&As and those involving foreign acquirers were significant in the long term but not the short term. In the context of trade firms, domestic M&As involving domestic acquirers and internal restructurings resulted in employment declines. However, there were no significant reductions in cross-border M&As and domestic M&As with foreign acquirers.
Hsieh & Davidson (2008); Link: 2c-3	Firms announcing one or more layoffs (1993-1995) in Lexis-Nexis. OLS regression using data from Compustat and Execucomp databases.	Outcome: Downsizing event Explanatory: Annual change in number of shares held by CEO	Increase in the number of shares held by the CEO was positively but marginally associated with employee layoffs.

2000) have found no significant relationship between firm performance and employee downsizing. Equivocal findings also characterize studies on the influence of market performance on downsizing. Although Budros (1997, 2000, 2004) observed that declining shareholder value induces downsizing, Hillier et al.'s (2006) study of layoffs in the United Kingdom surprisingly found a positive relationship between firm market value and employee reductions (suggesting that downsizing is often undertaken by firms with robust performance). Also, while a majority of the studies (Budros, 1997; Gittell, Cameron, Lim, & Rivas, 2006; Yoo & Mody, 2000) suggest a negative relationship between firm productivity and subsequent downsizing, dissenting studies by Hillier et al. (2006) and Yu and Park (2006) suggest that the link between employee productivity and downsizing is, at best, tenuous.

Studies belonging to Link 2a-3 have also examined the relationships between other firm attributes (e.g., size, age, market share, leverage, and reputation) and employee downsizing. Here, too, findings have been equivocal. For example, studies by Budros (1997), Coucke et al., (2007), Hallock (1998), Kang and Shivdasani (1997), and Wagar (1997) indicate that downsizing is more prevalent among larger firms. In contrast, Ahmadjian and Robinson (2001), Perry and Shivdasani (2005), Yoo and Mody (2000), and Yu and Park (2006) found no significant relationships between firm size and downsizing. Furthermore, while Ahmadjian and Robinson (2001) found that high sales growth reduces the incidence of downsizing, no significant effects were observed by Iqbal and Shetty (1994). Conflicting findings also characterize the relationships between firm leverage and employee downsizing. Although Gittell et al. (2006) and Hillier et al. (2006) provide evidence of a positive relationship between firm leverage and employee reductions, no such relationship was observed by Budros (1997) and Kang and Shivdasani (1997). There is, however, greater consensus in the context of market share, with most studies (Ahmadjian & Robinson, 2001; Budros, 1997, 2000) indicating that employee downsizing is often motivated by market share declines. Simultaneously, studies by Ahmadjian and Robinson (2001) and Coucke et al. (2007) found that layoffs were more common in younger firms. They attributed the lower incidence of downsizing in older firms to stronger inertial forces and less enthusiasm about downsizing. However, Budros (1997) found no support for the hypothesized relationship between firm age and downsizing. Finally, Ahmadjian and Robinson (2001) examined the effects of firm reputation on downsizing and, as expected, found downsizing to be more prevalent in firms having lower reputations.

In sum, as the above discussion highlights, research on the organizational antecedents of employee downsizing, although significant, is characterized by conflicting findings. The equivocal findings are indicative of the complexity of the underlying relationships and point to the importance of employing more complex (e.g., moderating and intervening effects) models in obtaining a better understanding of how such factors influence downsizing decisions.

Link 2b-3: Relationships between firm strategies and downsizing. A majority of the 11 studies categorized in this link have investigated the effects of M&As on subsequent employee downsizing. In these studies, downsizing is seen as an avenue by which firms eliminate redundancies resulting from M&As. Findings, however, have been mixed.

Specifically, whereas Budros (2002), Conyon, Girma, Thompson, and Wright (2002), and Lehto and Bockerman (2008) observed significant employment declines following M&As, studies by Budros (1997, 2004), and Wagar (1997) did not. In examining the relationship between M&As and downsizing, O'Shaughnessy and Flanagan (1998) observed that the relationship between the two was contingent on M&A type, with the likelihood of employee reductions being greater in the context of related transactions. Their findings were corroborated by Conyon et al. (2004). In addition, Conyon et al. (2004) found that hostile mergers were more likely to be accompanied by significant postmerger employment reductions than friendly transactions. Budros (2000) examined the moderating effects of industry type and found that downsizing was more likely in mergers undertaken in the manufacturing rather than the financial or utility sectors. Finally, in a study involving the effects of acquisition premiums, Krishnan, Hitt, and Park (2007) found that when acquiring firms paid higher acquisition premiums to target firm shareholders, it typically led to greater workforce reductions in the postacquisition phase.

Surprisingly, very limited research exists on the relationships between firm strategies (at the corporate or business level) and employee downsizing. Exceptions include relatively recent studies by Hillier et al. (2006) and Coucke et al. (2007), with the former concluding that the incidence of layoffs was higher among more diversified firms. Coucke et al. (2007), on the other hand, used a sample of Belgian firms to study the effects of competitive positioning on collective employee layoffs. They found that firms that emphasized product differentiation were less likely to engage in employee downsizing.

Link 2c-3: Relationships between firm governance, CEO characteristics, and downsizing. Studies belonging to this link have primarily used agency theory arguments in assessing the effects of governance mechanisms (board characteristics, ownership structures, and compensation systems) on employee downsizing. In an early attempt at understanding the relationships between ownership structures and downsizing, Bethel and Liebeskind (1993) found that firms with greater blockholder ownership experienced higher levels of employee reductions. In contrast, Filatotchev et al. (2000), in their study of layoffs among firms in Russia, Ukraine, and Belarus, observed no relationship between institutional ownership and downsizing. However, their study did indicate that although outside private ownership results in higher levels of employee downsizing, greater managerial ownership has the opposite effect. Similarly, a study by Stavrou, Kassinis, and Filotheou (2007) revealed that firms with high levels of family ownership engaged in less severe downsizing. Finally, in their assessment of the effects of foreign ownership on downsizing in Japanese firms, Ahmadjian and Robinson (2001) found that although such ownership resulted in increased likelihood of employee downsizing in a broad cross-section of firms, it was not so among nonfinancial firms. Research on the role of governance in the context of downsizing has also examined the role of board independence and board size in downsizing decisions. Findings suggest that firms with independent boards (i.e., a greater proportion of independent directors) are more likely to engage in downsizing (Perry & Shivdasani, 2005). However, unlike Perry and Shivdasani (2005), who found no significant relationship between board size and downsizing, Yawson (2006) observed that the propensity to downsize is lower among firms with larger boards.

A few studies in this link have focused on the effects of CEO demographic characteristics on downsizing decisions. In examining the role of CEO functional background, Budros (2000, 2002, 2004) found that the incidence of downsizing was higher in firms that had CEOs with a finance or operations background. However, this relationship was contingent on industry type, with a finance background being associated with downsizing only in the context of firms in the financial sector (but not the industrial or utility sectors). In addition, Budros (2002, 2004) observed a negative relationship between CEO age and downsizing, although this relationship was prevalent only among early adopters of downsizing. In contrast, Hallock (1998) found that neither CEO age nor tenure had any effect on the adoption of employee downsizing.

Link 2d-3: Relationships between HR practices, employee attributes, and downsizing. Most studies belonging to this link have explored the effects of firms' HR policies and practices on employee downsizing. Not surprisingly, Wagar (1997) found that when firms are more committed to employee job security, the likelihood of employee layoffs is diminished. However, he found no significant relationships between employee unionization and downsizing. In contrast, studies by Baumol et al. (2003) and Redman and Keithley (1998) provide evidence of a negative relationship between union density and employee downsizing. In addition, although it has been hypothesized that the level of employee compensation influences downsizing, a majority of the studies (Budros, 1997, 2000, 2001; Frazier, 2005; Iqbal & Shetty, 1994) indicate that employee compensation has no bearing on the extent or likelihood of employee downsizing. However, Yoo and Mody (2000), in their study of 50 U.S. local telephone companies, did find a positive relationship between mean employee compensation and the level of downsizing.

More recently, scholars have examined the effects of high commitment work practices (HCWP) on workforce reductions. Consistent with Osterman (2000), Iverson and Zatzick (2007) found that firms that make greater use of HCWP are more likely to engage in workforce downsizing. Their explanation for this finding is that in these workplaces, the use of HCWP tends to be part of a more general movement toward increased efficiency and productivity. Moreover, the use of these HR practices leads to a more competent, committed workforce and an ability to use leaner staffing. However, Iverson and Zatzick (2007: 471) also created a measure of downsizing "harshness" and found that greater use of high-commitment HR practices is associated with less harsh downsizing practices: "more HCWP is related to the use of downsizing, but it is through the use of more employee friendly strategies of voluntary layoffs and alternative strategies."

Stream III: Individual Outcomes of Employee Downsizing

Stream III research relates to the consequences of downsizing from the standpoint of employee attitude and behaviors. Downsizing has a significant potential to affect group and individual attitudes and behavior, disrupt relationship networks, and destroy the trust and loyalty that binds employees and their employers. The literature on organizational change (especially in the context of a significant event such as employee downsizing) suggests that

the consequence of change largely depends on how well the change process is managed. For example, it has been argued that effective communication, characterized by helpfulness (Miller & Monge, 1985), openness (Miller, Johnson, & Grau, 1994), and accuracy, timing, and completeness (Richardson & Denton, 1996), can help mitigate anxiety and reduce the resistance to change among downsizing survivors. Likewise, the change management literature recommends that the procedures used to manage change allow for employee inputs into the design of changes by providing them with opportunities to express their views and appeal decisions (Self et al., 2007). Enabling employees to maintain a sense of control should result in increased acceptance and commitment to the change (Cummings & Worley, 1993).

The influence of downsizing on individuals and groups in an organization has been examined in the context of psychological contract theory, which emphasizes the negative implications of violating implicit contracts governing relationships between firms and employees (Rousseau & Tijoriwala, 1998). Psychological contracts refer to perceptions about a set of mutual obligations that link employers and employees. They are based on expectations that each party will fulfill its obligation in accordance with their implicit contracts. The "contract" suggested in the downsizing literature is an expectation on the part of employees that their contributions to the workplace will be reciprocated with a stable and positive work environment. As such, downsizing is often perceived as a violation of the psychological contract (De Meuse, Bergmann, Vanderheiden, & Roraff, 2004; Morrison & Robinson, 1997; Rousseau, 1995) by survivors and victims alike. Given that contracts are fundamental to employees' beliefs and experiences, violations of such contracts have important consequences for employees' work attitudes and behaviors (Robinson, 1996). Surviving employees may withdraw psychologically (e.g., reduced trust and loyalty, withholding of effort, and reduced involvement) or physically (increased absences or voluntary turnover), with negative economic consequences for the firm.

Individual outcomes associated with downsizing have also been explained using justice theories, the basic tenet of which is that individuals evaluate situations with potentially important implications for them and react favorably if they perceive the situation to be "fair." Early work on distributive justice drew on equity theory to propose that individuals compare the ratio of their inputs and outcomes with those of relevant others to determine the fairness of their outcomes. Procedural justice (Thibault & Walker, 1975), on the other hand, concerns itself with the fairness of processes used in arriving at decisions in the allocation of resources (Hegtvedt & Markovsky, 1995). Processes are viewed as being fair when they provide for consistency across individuals and time, are free of bias, incorporate and reflect the opinions of people affected, and conform to moral and ethical standards (Leventhal, Karuza, & Fry, 1980). In the context of downsizing, perceptions of fairness related to downsizing decision-making and implementation processes have a significant bearing on subsequent behavior. Justice perceptions are influenced by employees' perceptions on whether layoffs were necessary, the appropriateness of the decision criteria used in identifying redundancies, and whether victims were fairly treated and adequately provided for after being downsized (Shah, 2000). If these are all deemed to be "fair" and "just," this evaluation can counteract and significantly offset the disappointment of downsizing. Conversely, perceptions of procedural *in*justice and *un*fairness are likely to magnify negative reactions and be manifested in reduced employee commitment and undesired behavioral outcomes.

The 26 studies in Stream III can be categorized into four groups based on the effects of downsizing on (a) job involvement, organizational commitment, loyalty, job satisfaction, and self-esteem (Link 3-4a with 20 studies); (b) trust, perceptions of justice, work conflict, and workplace environment (Link 3-4b with 10 studies); (c) voluntary turnover intent and absenteeism (Link 3-4c with 5 studies); and (d) work performance (Link 3-4d with 7 studies). Table 3 provides a summary of the empirical studies belonging to this stream.

Link 3-4a: Relationships between downsizing and job involvement, job satisfaction, loyalty, and self-esteem. Downsizing can have a profound impact on employee attitudes and behaviors, and these have been explored in the empirical literature associated with Link 3-4a. Although there is general agreement that downsizing results in lower job involvement and reduced organizational commitment among survivors (Allen, Freeman, Russell, Reizenstein, & Rentz, 2001; Brockner et al., 2004; Luthans & Sommer, 1999; Travaglione & Cross, 2006), studies by Armstrong-Stassen (1994) and Brockner et al. (2004) suggest that the effects are mitigated when downsizing is accompanied by high supervisor support and greater perceived control. Armstrong-Stassen's (1994) findings also suggest that job commitment following downsizing is a function of the coping strategy adopted by survivors with control coping (based on proactive goal setting and development of plans) resulting in greater commitment among survivors and escape coping (denial and distancing) having the opposite effect.

In addition to the above, studies by Spreitzer and Mishra (2002) and Chang (2002) indicate that negative effects of downsizing on the attachment and commitment of survivors is mitigated by management trustworthiness and perceptions of distributive justice (fairness of outcomes from downsizing). Spreitzer and Mishra's (2002) study also highlights the role of empowerment in increasing organizational attachment among downsizing survivors. On the other hand, they observed that low procedural justice led to significant reductions in self-esteem, especially among survivors with high organizational commitment.

Finally, Link 3-4a studies also indicate that downsizing results in reduced organizational commitment and job satisfaction (Gilson, Hurd, & Wagar, 2004; Luthans & Sommer, 1999; Travaglione & Cross, 2006). However, research by Allen et al. (2001) and Armstrong-Stassen (2002) suggests that adverse effects are relatively short-lived and diminish with time.

Link 3-4b: Relationships between downsizing, perceptions of trust and justice, and work environment changes. The limited research belonging to this link explores the effects of downsizing on organizational trust, perceived justice, workplace conflict, and changes in the work environment. The role of perceived justice in the context of downsizing was first highlighted by Brockner et al. (1994). They found that layoff severity did not have a perceptible impact on organizational trust and support when survivors perceived high procedural fairness. However, under conditions of low perceived procedural fairness, layoff severity had a significant deleterious effect on organizational trust. In a subsequent study, Mansour-Cole and Scott (1998) found that survivors perceive greater procedural fairness when they learn about impending layoffs from their managers and not from other sources. They also found that the relationship between the source of layoff information and perceived procedural

(Text continues on p. 321)

Table 3
Stream III: Individual Outcomes of Employee Downsizing

Study and Link	Sample	Variables	Findings
Brockner, Grover, & Blonder (1988); Link: 3c-4a	105 survivors in four organizations that had undergone downsizing. Employees (54) involved in mild layoffs were engineers and scientists in one organization. Severe layoffs (51) involved engineers in two engineering organizations. Ordinary least squares (OLS) regression of survey data.	Outcome: Job involvement Explanatory: Work ethic, role ambiguity, layoff condition (mild layoff group had experienced minor and sporadic layoffs in the 2 years preceding data collection; severe layoff group had experienced layoffs ranging from 25% to 70% in the 3 previous years)	In the mild layoff group, work ethic was positively associated with job involvement while prior role ambiguity was negatively associated with involvement. In the severe layoff group, work ethic and prior role ambiguity were not significantly associated with job involvement.
Mellor (1992); Links: 3a-4a, 3a-4a (3b)	355 union survivors after layoffs in 15 job sites in a single organization between 1980-1987. Moderated regression analysis of survey data.	Outcome: Union commitment (responsibility, loyalty, willingness to work for the union, belief in union) Explanatory: Layoff severity (% of layoffs at job site) Moderator: Belief in the account of the company regarding the layoff (255 in the low-belief group, 52 in the moderate-belief group, and 48 in the high-belief group)	Layoff severity was not significantly related to union commitment. Belief in the account of the company was negatively related to overall union commitment and also to loyalty ($p < .01$), willingness to work and belief in union. Interaction effects indicate that among survivors with a high level of belief in the account expressed less willingness to work for the union, while those with a lower level of belief in the account indicated greater willingness to work for the union. There was no relationship between layoff severity and willingness to work for the union among survivors with a moderate level of belief in the account.
Brockner, Grover, Reed, & DeWitt (1992); Link: 3c-4a	597 survivors of the downsizing in a U.S. retail chain with stores throughout the United States. Hierarchical regression analysis of survey data.	Outcome: Work effort Explanatory: Job insecurity (measured as perceived threat and perceived control), economic need to work	Job insecurity and economic need to work were not significantly associated with work effort. The perceived threat and perceived control interaction was positive and significant ($p < .05$), suggesting that work effort was greater when job security was medium (perceived threat and perceived control were

			either both high or both low) than under conditions of high or low job insecurity. For those with a high economic need to work, there was a significant inverted-U relationship between job insecurity and work effort ($p < .05$).
Brockner, Grover, O'Malley, Reed, & Glynn (1993); Link: 3b-4a (3c)	597 employees at a telecommunications retail firm in the United States that had undergone downsizing. OLS regression and ANOVA using survey and experimental data.	Outcome: Work motivation (before and after layoff condition) Explanatory: Threat of future layoff Moderator: Self-esteem, worry	The greater the threat of layoff, the lower was the motivation of survivors. The interaction between threat, self-esteem, and worry was positively associated with motivation, suggesting that high levels of worry among low-esteem survivors resulted in their working harder when the threat of future layoff was high.
Armstrong-Stassen (1994); Links: 3c-4a, 3c-4c, 3c-4d	146 technical employees of a major telecommunications firm in the United States. Hierarchical regression of data collected via surveys that were administered 6 weeks after a workforce reduction.	Outcome: Organizational commitment, turnover intention, self-reported job performance Explanatory: Coping resources—optimism, sense of mastery and perceived social support from coworker and supervisor, coping strategies (control, escape), stress appraisal (threat of job loss, sense of powerlessness) among downsizing survivors	Optimism had a negative impact and sense of mastery had a positive impact on turnover intention. Supervisor and coworker support were positively related to commitment. Supervisor support was negatively related to turnover intention and positively related to job performance. Threat of job loss was associated negatively with commitment and job performance and positively with turnover intention. Sense of powerlessness was negatively related to turnover intention. Control coping was positively related to commitment and job performance and negatively related to turnover intention; escape coping was

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Table 3 (continued)

Study and Link	Sample	Variables	Findings
			negatively associated with commitment and job performance and positively linked to turnover intention. In the context of high control coping strategy, the perceived threat of job loss was associated with negative turnover intention. In the context of escape coping strategy, a higher threat of perceived job loss was associated with greater turnover intention. Threat of layoff resulted in superior performance in the context of high control coping strategy. Among those using a high escape coping strategy, the job performance was lower when the threat of layoff was high.
Mishra & Mishra (1994); Link: 3a-4d	511 managers at 91 business units representing 43 North American automotive firms that had reduced employment in their operations in 1991. Interviews and OLS regression analysis of survey data.	Outcome: Product quality improvement, cost reduction, machine efficiency, and labor productivity improvements Explanatory: Workforce reduction	Workforce reduction had a negative impact on all aspects of quality improvement. It also had a negative impact on machine efficiency and labor productivity. There was no significant impact of workforce reduction on reduced material expenses.
Brockner et al. (1994); Links: 3a-4b, 3a-4b (3b)	3 studies with 218 layoff victims in one, 150 layoff survivors in the second, and 147 lame duck employees (to be laid off) in the third. OLS regression of survey data.	Outcome: Organizational trust and support Explanatory: Layoff severity (level of severance pay, extension of life insurance, extension of medical insurance coverage)	Layoff severity was not associated with organizational trust and support for all groups of employees. However, when perceived procedural fairness was low there was a negative relationship between layoff severity and organizational trust (in the victim, survivor and lame duck groups).
Mone (1994); Link: 3c-4c	145 employees at a U.S. manufacturing firm that experienced downsizing in the 1970s, 1980s, and 1990s. OLS regression of survey data.	Moderator: Procedural fairness Outcome: Survivor turnover intent Explanatory: Employee self-efficacy and self-esteem (global, role, and task self-esteem)	Self-efficacy was positively related to turnover intent. Task self-esteem, but not role and global self-esteem, had a negative effect on intent to leave.

Brockner et al. (1997); Links: 3b-4d, 3c-4d, 3c-4c, 3b-4c, 3b-4d	Study 1: 121 production worker-survivors at a unionized manufacturing facility that underwent downsizing 3 months earlier. Study 2: 597 downsizing survivors at a retail store chain in the United States. OLS regression of survey data.	Study 1: Outcome: Work performance Explanatory: Downsizing event, positivity of fellow survivors' reactions, perceived attractiveness of fellow survivors Study 2: Outcome: Work performance, turnover intention before and after downsizing Explanatory: Downsizing, positivity of fellow survivors' reactions, perceived attractiveness of fellow survivors, level of communication survivors had with fellow survivors about layoffs	Study 1: Among employees who were more attracted to their fellow survivors, work performance was greater when fellow survivors' reactions were perceived to be relatively positive. Survivors who were less attracted to their fellow survivors reported performing worse when their fellow survivors responded relatively positively to the layoffs. Study 2: When communication was low and attraction was high, survivors performed better when their fellow survivors responded positively; when both were low, work performance was worse when the reaction of other survivors was positive. When communication was high, survivors' work performance was positively related to other survivors' reactions, regardless of their attractiveness. When communication was low and attraction was high, survivors' turnover intention depended on how their fellow survivors responded. When communication was high, survivor's turnover intention was positively related to other survivors' reactions, regardless of their attractiveness.
Armstrong-Stassen (1998); Links: 3c-4a, 3c-4b	236 survivors (22 female clerical employees, 53 male technicians, and 39 female technicians (Group 1), 83 male technicians (Group 2), and 39 male first-level supervisors) of a division of a Fortune 100 U.S. corporation in the telecommunications industry.	Outcome: Perceived injustice, perceived job insecurity, coping strategies (positive thinking, direct action, help-seeking, and avoidance/resignation coping), emotional reactions Explanatory: Gender, organizational level (technicians and supervisors)	Female technicians reported greater procedural and distributive injustice related to the layoffs than did the male technicians. Female technicians engaged in more help-seeking coping than did the female clerical employees. Clerical employees were more likely to use direct action as a way of coping with layoffs than female technicians. Male technicians

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Table 3 (continued)

Study and Link	Sample	Variables	Findings
	MANCOVA and ANCOVA of survey data.		perceived greater procedural injustice related to the layoffs and reported a greater sense of powerlessness than did their male supervisors. Male supervisors engaged in more positive thinking and direct action coping than did male technicians, whereas the male technicians reported greater use of help seeking than did their supervisors. The group of technicians with greater exposure to workforce reductions perceived greater distributive injustice and reported higher threat of job loss than technicians with less exposure to reductions. They also reported greater use of positive thinking and direct action coping than did technicians.
Mansour-Cole & Scott (1998); Links: 3b-4d, 3b-4d (3c)	Survey of employees of the central R&D facility at a U.S. industrial corporation. 217 employees (21 managers, 108 engineers/scientists, 64 technicians, and 24 clerical employees) at time T1 (pre-layoff), 133 employees (13 managers, 61 engineers/scientists, 48 technicians, and 11 clerical employees) at time T2 (just after layoffs) and 78 employees (8 managers, 38 engineers/scientists, 24 technicians, and 8 clerical employees) at time T3 (2 years after layoff). Multiple regression and path analysis based on survey data.	Outcome: Distributive fairness (5 items measured at T2), procedural fairness (14 items measured at T2) Explanatory: Source of layoff announcement—single item measure, prelayoff and postlayoff affective commitment, legitimacy of organizational account of layoff Moderator: Extent of leader-member exchange (LMX)	The relationship between prelayoff affective commitment and procedural fairness was not significant. Survivors had higher procedural fairness perceptions when their managers personally informed them of impending layoffs than when informed from other sources. This effect was more pronounced for high LMX than for low LMX survivors. Also, the perceived legitimacy of the organizational account for a layoff was positively related to survivors' procedural fairness perceptions. Postlayoff affective commitment was positively related to procedural fairness perceptions but not to perceptions of distributive fairness measured after layoff.

Wagar (1998); Links: 3a-4a (2d), 3a-4b (2d)	1907 Canadian establishments with 55% reporting reductions in their workforce over a 2-year period in the 1990s. Ordered probit analysis of survey data.	Outcome: Employee satisfaction (employee morale, commitment to organization, employee satisfaction, and employee quality of life), employer-employee relations (composite of rate of grievances, absenteeism, employee turnover, employee resistance to change, and organizational conflict). Explanatory: Permanent workforce reduction (1 = *yes*, 0 = *no*), Size of workforce reduced (% of workforce reduced over a 2-year period) Moderator: High involvement workplace strategy	Permanent workforce reduction was associated with lower employee satisfaction and less favorable employer-employee relations. In establishments that had undergone workforce reduction, the adoption of high involvement workplace strategy was positively associated with employee satisfaction and more favorable employer–employee relations.
Amabile & Conti (1999); Links: 3a-4b, 3a-4d	754 employees at a Fortune 500 high-technology company undergoing downsizing (1993-1995). Groups of surviving employees sampled at four points in time: baseline (predownsizing, 1993), Wave 1 (middownsizing, July 1994), Wave 2 (end of downsizing, December 1994), and Wave 3 (postdownsizing, May 1995). 83, 23, and 26 employees were interviewed in waves 1, 2, and 3, respectively. MANOVA of survey data.	Outcome: Organizational creativity, organizational productivity, work environment (6 stimulants: organizational encouragement, sufficient resources, freedom, challenging work, supervisory encouragement, work group support; and organizational impediments: workload pressure) Explanatory: Downsizing	Creativity was lower than baseline levels at Wave 1, at Wave 2, and at Wave 3. Productivity declined from baseline levels at Wave 1 and Wave 2. During downsizing there was a decrease in work environment stimulants from baseline—freedom, work challenge, supervisory encouragement, work group support and resources available. On the other hand, organization impediments and workload pressure increased from baseline levels.

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Study and Link	Sample	Variables	Findings
Luthans & Sommer (1999); Links: 3a-4a 3a-4b	291 staff and managers at a 250 bed hospital in Midwest USA (1993-1995). ANCOVA with repeated measures of survey data.	Outcome: Organizational commitment, job satisfaction, supervisor support, workgroup trust Explanatory: Downsizing status, manager/staff (1 = *department downsized*, 0 = *otherwise*)	Downsizing led to reduced organizational commitment, job satisfaction, and workgroup trust. Both managers and staff employees reported significant declines in commitment, and employees scored lower on workgroup trust; managers reported high level on both commitment and workgroup trust throughout the downsizing.
Wiesenfield, Brockner, & Thibault (2000); Links: 3b-4a, 3b-4a (3c)	Study 1: 48 managerial and 81 nonmanagerial students who had survived a layoff. Study 2: 58 managers and 179 subordinates within 2 months of surviving an organizationwide downsizing. OLS regression of survey data.	Study 1:Outcome: Self-esteem Explanatory: Layoff procedural fairness during downsizing Moderators: Organizational commitment, managerial status (1 = *manager*, 0 = *otherwise*) Study 2: Outcome: Managerial behaviors Explanatory: Layoff procedural fairness Mediator: Manager's self-esteem	Study 1: Lower levels of layoff procedural fairness were associated with lower levels of self-esteem among employees with high organizational commitment and were associated with lower levels of self-esteem among managerial employees than among nonmanagerial employees. Managers' greater organizational commitment explains the more pronounced relationship between procedural unfairness and self-esteem. Study 2: Layoff procedural fairness was positively associated with managerial self-esteem and managerial behaviors. Managerial self-esteem mediates the relationship between procedural fairness and managerial behaviors.
Allen, Freeman, Russell, Reizenstein, & Rentz (2001); Links: 3a-4a, 3a-4b, 3a-4c	106 middle and upper level marketing-related managers at a large U.S. consumer packaged goods organization. Managers surveyed 1 month (T1), 4 months (T2), and 16 months after downsizing (T3) Repeated measured MANOVA; hierarchical regression.	Outcome: Employee attitudes (organizational commitment, role clarity, turnover intention, job involvement, satisfaction with top management) Explanatory: Downsizing	Organizational commitment, turnover intent, role overload, satisfaction with top management job security were more negative at T2 than at T1. However, there were no significant differences in turnover intent, job involvement, or satisfaction with job security between times T1 and T3, suggesting that attitudes were returning to their original level.

			Changes in role overload, clarity, satisfaction with top management, and job security were related to changes in organization commitment and turnover intention.
Armstrong-Stassen (2002); Links: 3a-4a, 3a-4d	49 employees who had been declared redundant and 118 who had not been designated to be downsized of a Canadian Government department in early 1996. Data collected in time T1 (early 1996), T2 (6 months later), T3 (1997) and T4 (1999). ANOVA of the 3-year survey longitudinal panel data.	Outcome: Job satisfaction, performance, perceived organization trust, perceived organization morale, organizational commitment, job security Explanatory: Redundant status (1 = *yes*, 0 = *no*)	Employees designated redundant reported significantly higher job satisfaction at T4 compared with the other three periods. While at T1, they reported lower job satisfaction than those declared nonredundant, but by T4 they reported higher job satisfaction than employees designated nonredundant. Both reported significant increase in job security between T1/T2 and T3 and also between T3 and T4. Survivors designated redundant reported greater trust in and commitment to the department by T4 than those not designated as redundant. Both groups reported significant increase in perceived organizational morale at T4. Overall, findings indicate that organizational downsizing does not have a long-term adverse effect on survivors.
Chang (2002); Links: 3b-4a, 3b-4a (3a)	411 employees working for 14 corporations in Seoul, Korea—162 at companies with layoffs, 249 in companies without. Moderated regression and ANOVA of survey data.	Outcome: Organizational commitment Explanatory: Distributive justice moderator: Layoff experience (yes/no), Layoff perception—likelihood of a layoff in the near future (0%, 1-20%, 21-40%, 41-60%, 61-80%, and 81-100%)	There was no significant interaction effect between distributive justice and layoff experience. A significant and positive interaction effect existed between distributive justice and layoff perception in explaining organization commitment. Layoff perception interacted with the perception of distributive justice by strengthening its positive effect of organizational commitment.

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Study and Link	Sample	Variables	Findings
Spreitzer & Mishra (2002); Links: 3b-4a, 3b-4c, 3b-3c-4a	350 full-time aerospace employees at a unionized plant undergoing downsizing. Downsizing announcement (that 10% of the site's contract workers were to be laid off) was made 1 month before surveys were distributed and turnover data collected a year later. OLS and logistic regression of survey and archival data.	Outcome: Organizational attachment (affective commitment), voluntary turnover (a year following downsizing) Explanatory: Downsizing, trust, justice—distributive, procedural, and interactional; Empowerment (four dimensions: meaning, competence, self-determination and impact) Mediator: Attachment (affective commitment)	Trustworthiness of management and perceptions of downsizing being distributively just were positively associated with organizational attachment. Higher commitment was also associated with procedural fairness. However, survivor assessment of interactional justice was not found to be related to commitment. The four dimensions of empowerment—meaning, competence, self-determination, and impact were significantly related to organizational attachment. Greater commitment among survivors resulted in lower likelihood of voluntary turnover. Survivor attachment to organization did not mediate the relationship between trust, justice, and empowerment and voluntary turnover.
Armstrong-Stassen, Wagar, & Cattaneo (2004); Links: 3a-4a, 3a-4b	159 nonmanagement survivors in a Federal Government department undergoing a mandated 20% reduction in its workforce in 1995. ANOVA using survey data.	Outcome: Change in job satisfaction (Job Diagnostic Survey), job security, job involvement, job performance, workload demands, interactional justice; change in group cohesiveness, service quality, supervisor emotional support, supervisor information support, employee morale Explanatory: Employee reduction, Change in work groups in past 2 years—intact (no new members), minor change (1 or 2 new members),	Survivors in minor change workgroup reported increased job performance and a significant decrease in perceived justice and job involvement. They reported a significant decrease in perceived informational support from their immediate supervisor. Survivors in moderate change groups reported increased job satisfaction and performance. They also reported increase in workload demands and increase in employee morale. Survivors in intact groups reported significant increase in perceived job security. They reported a significant decrease in perceived emotional support and

		moderate change (>25% of group were new), different group (respondents moved groups)	informational support from their immediate supervisor. Survivors who moved to a different group reported increased job satisfaction and perceived job security.
Brockner et al. (2004); Links: 3a-4a, 3a-4a (3b)	Study 1: Employees in two sites in the same division of an aerospace firm, one with (350 surveys) and the other without downsizing (787 surveys). Study 2: 103 survivors at a financial management company. Hierarchical regression analysis of survey data.	Study 1: Outcome: Organizational commitment Explanatory: Downsizing (1 = *with*, 0 = *without*) Moderator: Perceived control Study 2 Outcome: Postdownsizing performance evaluated by supervisor Explanatory: Threat to well-being Moderator: Perceived control	Study 1: Downsizing is negatively associated with organizational commitment. The negative impact of downsizing on survivors' organizational commitment is reduced when perceived control is high. Study 2: Survivors' postdownsizing job performance is negatively affected by threat to their well-being; however, the adverse effect is less pronounced when perceived control is high.
Gilson, Hurd, & Wagar (2004); Links: 3a-4a, 3a-4b	Downsizing in 322 New Zealand firms (1995-1999). Data collected from chief executive officers or managing directors. OLS regression based on panel data.	Outcome: Employee satisfaction, workplace conflict, workplace performance, employer performance Explanatory: Downsizing experience (repeat downsizers, early downsizers, late downsizers, and nondownsizers)	Late downsizers were associated with significantly lower employee satisfaction and workplace performance and higher workplace conflict when compared with nondownsizers. Compared with nondownsizers, repeat downsizers had significantly lower employee satisfaction, greater workplace conflict, and lower workplace performance. Compared with nondownsizers, early downsizers experienced marginally greater conflict and lower workplace performance.
Travaglione & Cross (2006); Links: 3a-4a, 3a-4b	141 employees in an Australian transport company undergoing downsizing in the 1990s. ANOVA using survey data.	Outcome: Employee organizational commitment (affective and continuance), perceived organizational support, job satisfaction (general and specific); employee job performance, turnover intention, and absenteeism Explanatory: Employee downsizing	Downsizing led to significant declines in affective commitment, effort, performance, and specific job satisfaction and to increased absenteeism. No significant change was observed in continuance commitment, perceived organizational support, general job satisfaction, or turnover intention.

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Table 3 (continued)

Study and Link	Sample	Variables	Findings
Armstrong-Stassen & Schlosser (2008); Links: 3c-4a, 3c-4d	237 managers during and following downsizing at two Canadian federal government departments targeted for a 20% or more reduction of their workforce. Data were collected at three time periods—18 months before downsizing (T1), during downsizing (T2), and 12 months after downsizing (T3). Hierarchical linear regression analysis based on survey data.	Outcome: Future success expectancy, job performance, job satisfaction, perceived coping effectiveness Explanatory: Generalized optimism Mediator: Future success expectancy at time T2, positive-thinking coping at time T2	Generalized optimism was positively related to survivors' future success expectancy during the downsizing and the use of positive thinking coping. Expectations for future success, measured during downsizing, mediated the positive influence of generalized optimism on expectations for future success, perceived coping effectiveness, job performance, and job satisfaction. Positive thinking coping mediated the positive relationship between generalized optimism (measured prior to downsizing) and expectations of future success, perceived coping effectiveness, job performance, and job satisfaction.
Brandes et al. (2008); Links: 3c-4a, 3c-4a (3c)	129 managerial employees at a national packaging company who survived a layoff 1 month earlier. Moderated regression analysis of survey data.	Outcome: Work effort Explanatory: Employee cynicism Moderator: Perceived job security among survivors	Perceived job security was negatively associated with work effort. Employee cynicism interacted with perceived job insecurity to determine postlayoff work effort, with work effort being highest when cynical layoff survivors concomitantly perceived high levels of job insecurity.
Trevor & Nyberg (2008); Links: 3a-4b, 3a-4a (2b)	106 and 161 firms in 1998 and 1999 respectively from the list of employee friendly companies to work for (Levering and Moskowitz, 2000). OLS analysis of archival and survey data.	Outcome: Voluntary turnover rate (number of voluntary leavers/number of full-time employees) Explanatory: Downsizing rate (% of employees downsized) Moderator: HR practices—procedural justice, job embeddedness, and career development	Downsizing rate was positively associated with voluntary turnover rate, and this relationship was mediated by organizational commitment. The positive effects of downsizing and voluntary turnover rate were lower in firms with HR practices associated with procedural justice perceptions and job embeddedness; the relationship was, however, stronger when firms had high levels of career development HR practices.

fairness is contingent on existing leader–subordinate relationships, with the association being more pronounced in relationships characterized by high mutual trust, respect, and obligation. In another study, Armstrong-Stassen (1998) observed significant differences in perceived procedural justice across different employee groups, with feelings of injustice being greater among female employees (relative to their male counterparts) and among technicians vis-à-vis their supervisors.

Link 3-4b studies also indicate that employee downsizing contributes to a decline in the postdownsizing work environment. Amabile and Conti (1999), for example, found that downsizing resulted in declines in work freedom, challenge, supervisory encouragement, work group support, and resource availability. Luthans and Sommer (1999), too, observed reduced supervisor support following downsizing. In addition, Gilson et al. (2004), in their study of downsizing among New Zealand firms, observed that firms that downsized were characterized by greater workplace conflict, and this relationship was consistent across early, repeat, and late downsizers.

Link 3-4c: Relationships between downsizing and turnover intent and absenteeism. Most of the studies belonging to this link have examined the effects of downsizing on the likelihood of voluntary turnover among survivors. Findings suggest that although downsizing increases the likelihood of such turnover (Allen et al., 2001; Trevor & Nyberg, 2008), the effects are less pronounced when employees have greater task self-esteem, are more optimistic, and/or exhibit greater organizational commitment (Armstrong-Stassen, 1994; Mone, 1994; Spreitzer & Mishra, 2002). In addition, Armstrong-Stassen's (1994) findings suggested that turnover intent is lower when employees feel a sense of powerlessness; with the opposite being true when they have a greater sense of mastery (i.e., they believe that they have greater control of their destiny) or when they perceive greater threat of job loss. Armstrong-Stassen (1994) also found that whereas control coping (active goal setting and development of plans) led to lower turnover intention, escape coping (denial and distancing) had the opposite effect. Interestingly, Allen et al.'s (2001) study indicated that turnover intent among survivors increased in the first 4 months following downsizing but subsided with time (the original baseline state was achieved by the 16th month following downsizing). Finally, recent work by Trevor and Nyberg (2008) reveals that the level of downsizing (the percentage of employees downsized) is positively associated with the extent of turnover, and this relationship is mediated by organizational commitment. They also found that procedurally just HR practices and job embeddedness reduced the extent to which downsizing led to turnover. Moreover, the downsizing–turnover relationship was greater among firms with higher levels of HR practices geared toward enhancing career development. Finally, there is some support for the notion that, in addition to increased turnover, downsizing results in increased absenteeism among survivors in the postdownsizing period (Travaglione & Cross, 2006).

Link 3-4d: Relationships between downsizing and work performance. The studies here primarily focus on the effects of downsizing on the quantity and quality of survivors' work. Studies indicate that downsizing results in reduced creativity (Amabile & Conti, 1999) and also has a negative impact on different aspects of quality improvement (Mishra & Mishra, 1994). Amabile and Conti (1999) attributed the reduced creativity to the deterioration in the

work environment following downsizing. In addition, there is some evidence that downsizing results in a significant decline in other dimensions of work performance (Amabile & Conti, 1999; Gilson et al., 2004; Mishra & Mishra, 1994; Travaglione & Cross, 2006). However, studies indicate that the relationship is influenced by several factors. Armstrong-Stassen (1994), for example, found that although supervisor support was positively associated with postdownsizing job performance, threat of job loss has an adverse effect on such performance. Likewise, Brockner et al. (2004) observed that the job performance of survivors following employee reductions was negatively affected by threats to their well-being. Moreover, unlike escape coping, control coping on the part of survivors had a positive impact on their job performance (Armstrong-Stassen, 1994). Finally, relationships with other survivors and their reactions played a key role in determining survivors' job performance, with greater attraction among survivors being associated with superior job performance. In addition, Brockner et al. (1997) found that communications coupled with positive reactions from fellow survivors resulted in better job performance among downsizing survivors.

Stream IV: Organizational Outcomes of Employee Downsizing

Much of the research belonging to this stream is geared toward addressing the basic question, "Does employee downsizing result in improved organizational performance?" Proponents of employee downsizing have long argued that downsizing represents a rational tool that managers can use to improve organizational productivity and efficiency (Cascio & Young, 2003). Improved firm performance can come from reduction of labor costs (which, in many organizations, is the largest component of the cost structure), provided reductions result in the remaining employees working at higher productivity levels. Critics of downsizing, on the other hand, have vehemently argued that such benefits are, at best, minimal and often nonexistent. They argue that although downsizing may result in short-term reductions in labor costs, it also undermines long-term competitive advantage by eroding skill bases, disrupting organizational relationship networks, and, as already discussed, eliciting negative responses on the part of survivors.

Although the literature belonging to this stream has been dominated by research on how downsizing affects financial performance (market returns and firm profitability), a few studies have examined other organizational outcomes, namely, sales growth, labor productivity, changes in R&D intensity, and reputation. The 36 studies here can be broadly categorized into three groups, namely, those studying downsizing outcomes from the perspective of (a) investor reactions and stock prices (Link 3-5a with 20 studies), (b) accounting returns (Link 3-5b with 19 studies), and (c) organizational efficiency and other outcomes (e.g., sales growth and R&D investments; Link 3-5c with 16 studies). The studies belonging to this stream are summarized in Table 4.

Link 3-5a: Relationships between employee downsizing and market returns. Our review identified 12 studies that have used the event study methodology to examine shareholder wealth effects associated with employee downsizing announcements. The wealth effects have been typically measured as cumulative abnormal returns (CARs) or cumulative prediction errors (CPEs) around the announcement day. Examination of research belonging to this link

(Text continues on p. 335)

Table 4
Stream IV: Organizational Outcomes of Employee Downsizing

Study and Link	Sample	Variables	Findings
Worrell, Davidson, & Sharma (1991); Link: 3a-5a	194 layoff announcements (1979-1987) identified in *Wall Street Journal* (*WSJ*). Event study and subgroup analysis using data from Compustat and Center for Research in Security Prices (CRSP).	Outcome: Cumulative abnormal returns (CARs) for the (–5, +5) and (–1, +1) windows Explanatory: Layoff reason (financial, restructuring, consolidation), layoff size, layoff duration	Layoff announcements resulted in significant negative CARs in the (–5, +5) window. CARs associated with permanent layoffs and layoffs because of financial distress were significantly more negative that those resulting from restructuring. There were no significant differences in CARs between large (>4.5%) and small (<.98%) layoffs.
Linn & Rozeff (1993); Link: 3a-5a	246 temporary and 137 permanent layoff announcements (1978-1985) reported in the *WSJ*. Event study using market model and stock data from CRSP.	Outcome: Cumulative prediction errors (CPEs) for the 2-day (–1, 0) and the post-announcement (1, +60) window Explanatory: Temporary and permanent layoffs of (a) hourly and (b) salaried labor	Temporary layoffs of hourly and salaried labor and permanent layoffs of hourly labor were associated with negative CPEs for the (–1, 0) window, but not for the (1, +60) window. Permanent layoff of salaried labor was associated with significant negative CPEs for both the (1, 0) and the (1, +60) period.
De Meuse Vanderheiden, & Bergmann (1994); Links: 3a-5a, 3a-5b, 5c	17 Fortune 100 firms with layoff announcements in 1989—data from the Forbes annual survey of Fortune 500 firms. OLS regression and subgroup analysis.	Outcome: Profit margin/sales, return on asset (ROA), return on equity (ROE), asset turnover, market-to-book ratio Explanatory: Percentage layoff (layoff as a % of total employees)	Compared with control firms, layoff firms in the first and second years following layoffs had lower profit margin on sales, ROA, ROE, and market-to-book ratio. No significant difference was observed for asset efficiency.
Chatrath, Ramchander, & Song (1995); Link: 3a-5a	231 layoff announcements (1981-1992) reported in *WSJ*. Event study using market model and archival data from CRSP tapes.	Outcome: CPEs in the (–10, +10) and (0, +1) windows. Explanatory: Layoff announcements during (a) 1981-1983 (35), (b) 1984-1990 (56) and (c) 1991-1992 (140)	Layoff announcements were associated with negative CPE in the (0, +1) window in the periods 1981-1983 and 1984-1990; however, CPE was positive in the 1991-1992 time period. CPEs for (–10, +10) were significant and negative only in the 1984-1990 period.
Dougherty & Bowman (1995); Link: 3a-5c	106 people working on new products at 6 high and 6 low downsizing firms.	Outcome: Effectiveness of product innovation (product conceptualization, lateral integration and strategic linking)	Downsizing hindered product innovation and strategic problem solving by breaking down the network of informal relationships used to work

(continued)

Table 4 (continued)

Study and Link	Sample	Variables	Findings
	Univariate *t* tests.	Explanatory: High or low downsizing firms (based on how many times downsizing or reorganization was mentioned in the letter to shareholders in the annual report)	out strategic linkages. There was no relationship between downsizing and product conceptualization or lateral linking problem solving. Innovators in high downsizing firms solved significantly fewer strategic linking problems (linking products to firm's resources, structure and strategy) than low downsizing firms.
Bruton, Keels, & Shook (1996); Links: 3a-5b, 3a-5d	100 Fortune 500 companies that reduced workforce by at least 3% (1985-1987). Regression analysis using data from Compustat.	Outcome: ROA change, change in R&D spending Explanatory: Workforce reduction; prior financial health	Extent of workforce reduction was not a factor in explaining postdownsizing ROA. Workforce reductions produce benefits regardless of the firm's prior financial health. Downsized firms reduced their R&D spending, which improved financial performance that persisted for 4 years after downsizing.
Kang & Shivdasani (1997); Link: 3a-5b	92 Japanese firms compared with 114 U.S. firms that experienced significant performance declines (1986-1990). Layoff data collected using *WSJ*, Dow Jones News Retrieval Service, and Lexis-Nexis. Subgroup analysis using archival data.	Outcome: Change in operating performance (operating income to assets) over 1, 2 and 3 years Explanatory: Layoffs	There were no significant changes in operating performance in the first and second year following layoffs. Improved operating performance resulted only in the third year following layoffs.
Cascio, Young, & Morris (1997); Links: 3a-5a, 3a-5b	722 employee downsizing occurrences in 537 companies (1980-1994). Archival data from Compustat.	Outcome: Change in ROA between year prior to downsizing and 0, 1, 2 years (0 = downsizing year); Stock returns in 0, 1, 2 years Explanatory: Employment downsizers (employment decline > 5% but < 5% in assets); stable employers (less than 5% change in both employment and assets)	Change in ROA for employment downsizers was negative in both the downsizing year and the following year. It was also significantly worse than for stable employers. Employment downsizers had a significantly lower mean stock return than stable employers in the year of downsizing; however, such returns were significantly higher than stable employers in the year following downsizing.

Palmon, Sun, & Tang (1997); Links: 3a-5a, 3a-5b, 3a-5c	140 layoff announcements 1982-1990 (57 citing declining demand and 83 efficiency considerations) reported in *WSJ* and *New York Times* (*NYT*). Event study and subgroup analysis using market model and data from CRSP and Compustat databases.	Outcome: CARs in (–1, 0), (–1, +1) and (–1, +10) day windows, profit margin (income/sales), ROA ROE, real sales (sales deflated by the producer price index) for the years –3 to +3 with "0" being the layoff year. Explanatory: Layoffs motivation (a) declining demand and (b) efficiency enhancement	Layoffs motivated by declining demand were associated with negative CARs in the (–1, 0), (–1, +1) and (–1, 10) day windows, while those motivated by efficiency enhancement had positive CARs in the (–1, 0) and (–1, +1) day windows. Declining demand layoffs did not have a significant impact on profit margin, ROA, ROE, or real sales. Efficiency enhancing layoffs were associated with increased profit margins between Years –1 and +3 and Years –2 and +3, increased ROE between Years –1 and +2, increased ROA between Years –2 and +2 and between Years –2 and +3. Such layoffs also resulted in increased real sales between Years –2 and +2.
Lee (1997); Link: 3a-5a	300 U.S. layoff announcements and 73 in Japanese firms (1980-1994) reported in *WSJ*, Asian *WSJ*, *Nihon Keizai Shimbun*, and other sources. Event study and regression analysis using data from various sources.	Outcome: CARs over the 5-day window (–2, +2) surrounding layoff announcements Explanatory: Crossholdings, layoff characteristic—temporary or permanent, layoff size (percentage laid off); proactive and reactive layoffs, first or subsequent layoff in a given industry	Layoff announcements were associated with negative CARs for both U.S. and Japanese firms with CARs for the Japanese sample being significantly less negative than the U.S. sample. CARs were negative for permanent layoffs but not for temporary layoffs in U.S. firms. They were also negative for reactive layoffs but not for proactive layoffs in the U.S. In Japan, multiple layoff announcements were associated with negative CARs but not single announcements. Both single and multiple announcements were associated with significant negative CARs in U.S. firms. There was also a curvilinear relationship between the percentage of work force laid off and the CARs among U.S. firms.

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Table 4 (continued)

Study and Link	Sample	Variables	Findings
Elayan, Swales, Maris, & Scott (1998); Links: 3a-5a, 3a-5a (1b), 3a-5a (2a)	646 layoff announcements between 1979-1991 identified using the *WSJ* and the National Newspaper Index. Event study and subgroup analysis using data from CRSP and Compustat databases.	Outcome: CARs over the (-1, 0) window Explanatory: Size of layoff, duration of layoff (permanent vs. temporary), anticipated vs. unanticipated layoff, reason for layoff (operations not profitable, restructuring, labor-management dispute, operations closing), industry classification, business cycle (expansion or contraction), ROE, net income/employee, sales/employee	CAR for the 2-day window (–1, 0) was negative. Firms with ROE, net income/employee and sales/employee greater that the industry average experienced significant negative CARs in the (–1, 0) period, while those for firms with returns below industry average were not significant. Layoffs both above and below the median layoff size were associated with negative CARs. CARs in the context of permanent layoffs were negative and insignificant for temporary layoffs. Those associated with unanticipated announcements were also negative, as were CARs associated with layoffs caused by unprofitable operations. However, those associated with restructuring and discontinued operations were not significant. Layoffs associated labor management disputes resulted in negative CARs. CARs associated with layoffs in manufactured process product industries were significant and positive. In the context of business cycles, layoffs in both expansion and contraction periods resulted in negative CARs.
Franz, Crawford, & Dwyer (1998); Links: 3a-5a, 3a-5a (2a)	351 involuntary layoff announcements (1978-1992) reported in *WSJ*. Event study and subgroups analysis using data from CRSP and Compustat.	Outcome: CARs for the (–1, 0) and (–5, +5) windows Explanatory: Motivation for downsizing (cost reduction and reorganization); financial condition of downsizing firms—distressed, marginal and healthy (using Altman's Z scores)	Downsizing was associated with negative CARs for both the (–1, 0) and (–5, +5) windows (when downsizing was undertaken for cost reduction but not for reorganization). The 2-day (–1, 0) CARs for distressed, marginal, and healthy firms were all negative and significant; however, healthy firms that downsized had the highest negative CARs.

Hallock (1998); Link: 3a-5a	550 U.S. firms with layoff announcements (1987-1995) reported in *WSJ*. Archival event study and subgroup analysis using data from CRSP and Compustat.	Outcome: CARs for the (–1, +1), (–5, +5), (–10, +10) and (-20, +20) windows Explanatory: Permanent and temporary layoffs	Permanent layoffs were associated with significant negative CARs for (–1, +1), (–10, +10), and (–20, +20) windows. However, CARs associated with temporary layoffs were not significant except in the (–20, +20) window, where they were negative and marginally significant.
Wagar (1998); Link: 3a-5c	1907 Canadian establishments with 55% reporting reductions in their workforce over a 2-year period in the 1990s. Ordered probit analysis of survey data.	Outcome: Employer efficiency (combination of productivity, product quality, and customer satisfaction) Explanatory: Permanent workforce reduction (1 = *yes*, 0 = *no*), size of workforce reduced (% of workforce reduced over a 2-year period)	Permanent workforce reduction was associated with lower employer efficiency.
Espahbodi, John, & Vasudevan (2000); Links: 3a-5b, 3a-5c	118 firms that announced employee downsizing (1989-1993) in *WSJ* or *NYT*. Analysis comparing downsized and control firms using data from Compustat using parametric *t* tests and Wilcoxon signed-rank test.	Outcome: Raw operating performance, industry adjusted operating performance, matching firm adjusted operating performance, ROA, ROE, current ratio, total asset turnover, changes in labor costs as a % of sales, R&D as a % of sales, cost of sales/sales Explanatory: Downsizing announcements	Downsizing resulted in increased raw and industry adjusted operating performance and also matched-firm adjusted performance (between year prior to downsizing and the third and fourth year after downsizing, respectively). Downsized firms relative to control firms (a) exhibited higher ROE (but not ROA), (b) had a lower current ratio, and (c) had reduced firm R&D and advertising intensity in the third and fourth years following downsizing.
Wayhan & Werner (2000); Links: 3a-5a, 3a-5d	225 large U.S. firms (1991-1992) divided into two groups based on whether they announced a workforce reduction initiative of at least 3%. MANOVA with repeated measures using archival data from S&P stock	Outcome: Growth in sales and market capitalization over 6 time periods (1989-1991, 1990-1992, 1991-1993, 1992-1994, 1993-1995, and 1994-1996) Explanatory: Downsizing (at least 3% reduction in workforce)	Workforce reductions resulted in the reduction of sales growth performance gap that existed prior to downsizing, with the difference becoming nonsignificant in the second (1991-1993) and third (1992-1994) postworkforce reducing periods. However, the gap returned to significant differences in the last two time periods.

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Table 4 (continued)

Study and Link	Sample	Variables	Findings
	market data and Fortune 500 market listings.		Results for growth in market capitalization showed a turnaround in performance for companies with a significant gap in market capitalization growth rate prior to workforce reduction. The gap was totally eliminated by the first post–workforce reducing time period and marginally significant in the opposite direction during the second postdownsizing time period.
Chen, Mehrotra, Sivakumar, & Yu (2001); Links: 3a-5a, 3a-5b, 3a-5c	349 layoff announcements (1990-1995) reported in the *WSJ*. Archival data from CRSP and Compustat. Wilcoxon signed-rank test.	Outcome: CARs for the (-1, 0) window, changes in ROA, operating earnings/sales, cost of goods sold/sales, sales and administrative expenses/sales, sales/employee, capital expenditure/employee Explanatory: Layoff vs. nonlayoff firms	Downsizing announcement was associated with significant negative CARs of -1.30% in the 2-day (–1, 0) period. ROA, operating earnings/sales and sales/employee for layoff firm median was greater than that of the industry adjusted mean for the periods 0 to 3 years after layoff. Cost of goods sold/sales was lower for layoff firms. No significant differences were observed for sales & administrative expenses/employee. Relative to nonlayoff firms, layoff firms had greater improvements in both ROA and operating margins/sales in the 3-year postlayoff period. Labor productivity (sales/employee) in layoff firms increased faster than those in nonlayoff firms.
McElroy, Morrow, & Rude (2001); Link: 3a-5b	31 subunits of a national financial services company. Survey and organizational record data. Partial correlation analysis.	Outcome: Performance (basis points of profit/loan) over 2 years (year of downsizing—Year 1 and the following year—Year 2) Explanatory: Employee reduction (i.e., dismissals, voluntary turnover, and reduction in force turnover)	Employee reductions brought about by redundancy had significant negative performance consequences that extended into Year 2. Involuntary turnover (dismissals) was negatively related to customer satisfaction and to Year 2 costs per loan. Reduction in force turnover was

			negatively related to Year 1 profitability, customer satisfaction, Year 2 profitability, Year 2 productivity, and cost per loan.
Suárez-González (2001); Links: 3a-5b, 3a-5c	297 large Spanish firms (141 downsizers and 156 nondownsizers). Archival data from *Fomento de la Producción* and from Dun and Bradstreet.	Outcome: Labor productivity 1994-1996 (sales/employee), Return on sales 1994-1996 Explanatory: Employee downsizing (reduction of workforce by at least 5% between 1989-1994)	Downsizers had significantly lower return on sales in the postdownsizing period than non–downsizing firms. There were no significant differences between downsizers and nondownsizers in the labor productivity following downsizing.
Chalos & Chen (2002); Links: 3a-5a, 3a-5b, 3a-5c	Univariate *t* tests. Employee downsizing in 365 Fortune 500 firms (1993-1995) identified using *WSJ*. Event study using market model and regression analysis based on data from CRSP and Compustat.	Outcome: CARs on Day 0 and (-1, +1), change in ROA, sales productivity, cost of goods sold, efficiency, operating cash flow per employee Explanatory: Proportion of downsized employees (employees downsized/total number of employees), Type of downsizing (revenue refocusing and cost cutting)	Revenue refocusing downsizing announcements were associated with positive CARs on Day 0 but not in cost-cutting downsizings. Revenue-refocusing and cost-cutting downsizings both resulted in significant increases in ROA, sales productivity, and operating cash flow per employee, but not in cost-of-goods-sold efficiency. Proportion of downsized employees was not associated with CARs for (-1, +1); however, CARs for downsizing announcements of revenue refocusing were significantly more positive than those of cost cutting announcements.
Baumol, Blinder, & Wolff (2003); Links: 3a-5a, 3a-5b, 3a-5c	267 establishments in 20 manufacturing industries (1967-1997). OLS regression of secondary data from National Income and Product Accounts, Bureau of Labor Statistics, Department of Commerce, and BEA.	Outcome: Productivity growth, profitability, market value, employee compensation, unit labor cost Explanatory: Downsizing measured in different time periods as the percentage change in mean employment per establishment and percentage change in mean employment per firm.	There was no direct association between downsizing and total factor productivity growth. Downsizing at both the establishment and firm level had a positive effect on profitability. However, employee reduction was associated with decreased market value of the firm. Downsizing had a negative impact on growth in employee compensation and also led to lower per-unit costs.

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Table 4 (continued)

Study and Link	Sample	Variables	Findings
Cascio & Young (2003); Links: 3a-5a, 3a-5b	657 employment change occurrences (1982-2000). Analysis using Compustat data.	Outcome: Change in ROA between base year $t = -1$ to years $t = 0$, 1, 2, and stock returns in years 0, 1, and 2 (where 0 was the downsizing year) Explanatory: Employment downsizing (companies where the decline in employment was greater than 5%)	Relative to base year, the ROA of employment downsizers declined in Years 0 and 1 and rose slightly in Year 2. However, by the end of Year 2, the change in ROA for employment downsizers was lower than stable employers. Employment downsizers had a significantly lower mean stock return than stable employers in the year of the downsizing. There were no significant differences between the two in Years 1 and 2 following downsizing.
De Meuse, Bergmann, Vanderheiden, & Roraff (2004); Links: 3a-5a, 3a-5b	Layoff announcements by 78 Fortune 100 firms (1989-1996) reported in *Work Place Trends* and *WSJ*. Comparison of layoff and no layoff firms using financial data from *Fortune Survey* and *Forbes 500*.	Outcome: Profit margin (profits/sales), ROA, ROE, asset efficiency (sales/assets), market-to-book ratio Explanatory: Extent of downsizing (announced reductions divided by the total number of employees in the company) during 1989-1996	Compared with non–layoff firms, those that engaged in layoffs had significantly lower profit margins, market to book ratios, ROA, and ROE in the announcement year and in the 2 subsequent years. There were no significant performance differences between firms that laid off less than 3% and those that laid off 3% or more employees. Firms that laid off 10% or more underperformed those that laid off a lower percentage of employees along profit margins, ROA, ROE, and market to book in the 3 years following layoffs.
Nixon, Hitt, Lee, & Jeong (2004); Links: 3a-5a, 3a-5a (2d)	364 downsizing announcements by US firms (1990-1998) identified using *WSJ* and Lexis-Nexis. Event study and generalized least squares (GLS) regression analysis of data from CRSP and Compustat databases.	Outcome: CARs for the 3-day (–1, +1) window Explanatory: Level of downsizing (% reduction in total employees in a firm). Moderator: *High reallocation downsizing* = 1; *Low reallocation (across the board) downsizing* = 0, disengagement incentives: 1 = *offered* and 0 = *if not offered.*	Downsizing resulted in negative CARs, with negative effects being more pronounced at higher levels of downsizing. Reallocation level in downsizing was positively associated with CARs. Reallocation positively moderated the relationship between the level of downsizing and market valuation of the firm. The announcement of disengagement incentives positively moderated the relationship between a firm's level of downsizing and market valuation of the firm.

Chadwick, Hunter, & Walston (2004); Links: 3a-5b, 3a-5b (3b)	114 acute care major hospitals. Ordered logit and OLS estimations using survey data from hospital CEOs and other upper managers.	Outcome: Managerial perception of downsizing success, cash margin (ratio of free cash available for a hospital's use to the total hospital net revenues) Explanatory: Workforce downsizing (workforce reduction of at least 15%), Strategic HR practices: (a) advance notice given to downsized employees, (b) benefits offered after downsizing in the form of extended insurance coverage (1 = *yes*, 0 = *no*); (c) planned redesign of work structures; (d) consideration for employee morale	Consideration for employees was positively associated with perceived success of the downsizing initiative. However, the effect on cash margin was not significant. More extensive advance notice or benefits coverage had no significant effect on perceived success of downsizing. More advanced notice was significantly and positively related to cash margin in the year of downsizing. Provision of extended insurance coverage to laid-off employees was negatively associated with cash margin ($p <$ .05). The effects of consideration for employees and advance notice emerged as significant over time. Planned redesign was positively associated with perceived success but had a neutral-to-negative effect on cash margin.
Love & Nohria (2005); Links: 3a-5b, 3a-5b (2a)	Downsizing in Fortune 100 firms (1977-1993) identified using on *WSJ* and *NYT* and several wire services. Pooled time series regression analysis using Compustat data.	Outcome: ROA market (based on market valued assets), ROA book (based on book valued assets) Explanatory: Downsizing Moderator: Absolute slack (SG&A expenses) and relative absorbed slack (SG&A expenses relative to similar firms), downsizing scope (broad = *structural/process changes and changes in strategic domain of firm*; narrow = *otherwise*), downsizing timing (proactive or reactive)	Downsizing did not directly act on postdownsizing performance. High absorbed-slack firms exhibited better postdownsizing performance than low-absorbed-slack firms. Relative absorbed slack is not significantly associated with postdownsizing performance. Broadly scoped downsizings were associated with better postdownsizing performance than narrowly scoped downsizings. Proactive downsizings experienced better postdownsizing performance than reactive downsizings.
Zyglidopoulos (2005); Link: 3a-5d	46 firms belonging to Fortune's "America's Most Admired Corporations" list (1988-1991). *WSJ* Index used to identify firms that engaged in downsizing.	Outcome: Overall reputational score Explanatory: Employee downsizing—operationalized as (a) any reduction in employee headcount, (b) as any reduction in employee head count > 5%.	Employment downsizing was associated with reduced corporate reputation scores ($p < .01$ in the context of any reduction in employees and $p < .001$ for downsizing involving employee reductions greater than 5%).

(continued)

Table 4 (continued)

Study and Link	Sample	Variables	Findings
	Regression analysis using data from the AMAC database.		
Perry & Shivdasani (2005); Links: 3a-5b, 3a-5b, 3a-5b (2c)	94 nonfinancial firms in the U.S. that downsized in 1993 after financial decline (at least 33% drop in pretax income). Analysis conducted using archival data from Compustat, Execucomp database, *WSJ*, Dow Jones Retrieval Service.	Outcome: ROA change between the year of employee reduction and the following 2 years Explanatory: Employee downsizing Moderator: Outside boards	There were no significant ROA changes in firms that engaged in employee reduction between the year of downsizing and the subsequent 2 years. Among firms with outside boards that reduced employment, there was a significant positive change in ROA between the year of employment reduction and 2 years following the reduction. Employment reductions made in firms with inside boards had a negative but nonsignificant change in ROA for the same period.
Flanagan & O'Shaughnessy (2005); Links: 3a-5d, 3a-5d (2a)	347 firms in the (1996-1998) Fortune AMAC list. Layoff announcements identified based on *WSJ*. OLS regression of archival data from AMAC Survey database and Compustat.	Outcome: Firm reputation Explanatory: Layoff (1 = *layoff*, 0 = *otherwise*) Moderator: Prior performance (prelayoff industry adjusted ROA), relayoff organizational size (log industry-adjusted sales), firm age	Layoffs had a significant negative effect on firm reputation. Prelayoff ROA or firm sales did not moderate the relationship between layoff and reputation. The negative effect of layoff on reputation was significantly stronger in newer firms than older firms. The moderating effects of firm size on the relationships between layoffs and overall reputation were marginally significant.
Yu & Park (2006); Links: 3a-5b, 3a-5c, 3a-5b (2a)	258 publicly traded firms in Korea (1997-1999). OLS regression of survey data and archival firm data from KIS-FAS database.	Outcome: ROA change between pre- and postdownsizing periods, asset turnover (sales/ average total assets), operating income/ employee, sales/ employee, value-added/ employee Explanatory: Downsizing Moderator: Loss status (1 = loss in the 3 years prior to downsizing; 0 = otherwise)	Downsizing was positively associated with ROA, asset turnover, and operating income to total assets. However, there were no effects on productivity (neither sales/employee nor value added per employee). Downsizing resulted in a positive and significant improvement in ROA among firms that had no losses in the 3 years prior to downsizing. No significant relationships were observed in the context of other measures for either loss-making or non–loss making firms.

Zatzick & Iverson (2006); Links: 3a-5c, 3a-5c (2d)	3,080 Canadian workplaces in 1999-2001 and 2,970 workplaces in 2002. Moderated hierarchical regression analysis using data from Statistics Canada Workplace Employment Survey.	Outcome: 2001 and 2002 employee productivity (log of revenues-expenditures per employee) Explanatory: Employee layoff rate (layoffs/number of employees) during 2000-2001 Moderator: High involvement work practices (HIWP; based on five practices)	In firms making more extensive use of HIWP the negative effect of layoffs on productivity was stronger than in firms with less extensive use of high involvement work practices. Firms that maintain their investments in HIWP following layoffs were able to stem productivity losses.
Brookman, Chang, & Rennie (2007); Link: 3a-5a	Layoff announcements by 484 firms (1993-2003) reported in *WSJ*. Event study using market model and data drawn from CRSP and OLS regression using data from Compustat, CRSP and ExecuComp.	Outcome: CARs in the 6-day (–5, 0) and 2-day (–1, 0) windows Explanatory: Layoff announcements, layoff size (% of total workforce laid off)	Layoff announcements resulted in significant positive CARs in the 6-day (–5, 0) window but not in the 2-day (-1, 0) window. CARs for the 2-day window were not significant. There was a positive relationship between layoff size and layoff-related CARs in the (-5, 0) window.
Said, Le Louarn, & Tremblay (2007); Links: 3a-5c, 3a-5b	140 downsizers and 99 stable employers in the U.S. and Canadian manufacturing sector (1990-1996). ANOVA with repeated measures using data from Compustat.	Outcome: Labor productivity (log of sales/employees), operational indebtedness (liabilities/total assets) in the 3 years following downsizing Explanatory: Employee downsizing (at least 5% employee reduction)	Compared with stable employers, employee downsizers did not show any significant change in their labor productivity and indebtedness ratio in the short and medium term. Firms that engaged in large employee cuts (>20% of workforce) reported a significant increase in operational indebtedness in the 3 years following downsizing.
Hillier, Marshall, McColgan, & Werema (2007); Link: 3a-5a	322 layoff announcements by U.K. firms (1990-2000). Univariate analysis using data from the LSE, FT Extel, Lexis-Nexis, and Datastream.	Outcome: CARs in the 3-day (–1, +1) window Explanatory: Layoff, stated reason for layoff, size of layoff (greater or less than 4.1%)	Layoff announcements were associated with negative CARs in the 3-day (–1, +1) window. Layoffs brought about by plant closures and loss-making activities were associated with significant negative CARs. Layoffs greater than 4.1% were associated with higher negative CARs.

(continued)

Table 4 (continued)

Study and Link	Sample	Variables	Findings
Guthrie & Datta (2008); Links: 3a-5b, 3a-5b (1b)	144 nondiversified U.S. firms (1998-2000). Survey data from HR managers in these firms. OLS regression using firm and industry data from Census and Compustat.	Outcome: ROA Explanatory: Employee downsizing measured as a dichotomous variable (1 = *if employment reduction exceeded 5% between 1998 and 1999,* 0 = *otherwise*). Moderator: Industry R&D intensity, demand instability, capital intensity and growth	There was a negative association between workforce downsizing and postdownsizing firm ROA. The negative effect of downsizing on performance was magnified by industry R&D intensity and industry growth. The negative relationship between downsizing and firm performance was also more pronounced in industries with lower capital intensity. No significant moderating effect was observed for industry demand instability.
Goins & Gruca (2008); Link: 3a-5a	71 layoff events (1989-1996) among oil and gas companies. Event study methodology and OLS regression using archival data.	Outcome: Cumulative abnormal returns in the (–1, 0) and (–5, +5) windows Explanatory: Permanent layoff announcements involving at least 1% of employees	CARs for the (-1, 0) and (-5, +5) windows were negative but not significant.

reveals that, on average, downsizing announcements have a negative effect on stock price. A majority (9 of the 12) of the studies (Chatrath, Ramchander, & Song, 1995; Chen, Mehrotra, Sivakumar, & Yu, 2001; Elayan, Swales, Maris, & Scott, 1998; Franz, Crawford, & Dwyer, 1998; Hallock, 1998; Hillier et al., 2006; Lee, 1997; Linn & Rozeff, 1993; Nixon, Hitt, Lee, & Jeong, 2004; Worrell, Davidson, & Sharma, 1991) found that downsizing announcements resulted in negative wealth effects. In other words, investors, on average, react negatively to news of employee downsizing. Dissenting studies include that by Brookman, Chang, and Rennie (2007), who found downsizing announcements to be associated with positive returns in the (−5, 0) window, and Chalos and Chen (2002), who observed that downsizing was associated with positive CARs on the announcement day. In a more recent study, Goins and Gruca (2008) found no significant shareholder returns in the period surrounding the layoff announcement. Studies also indicate that market responses depend on the nature of the downsizing decision. For example, Palmon, Sun, and Tang (1997) found that wealth effects were contingent on downsizing motives and goals, with positive returns in situations where layoffs were initiated with the objective of improving efficiency, but not when layoffs occurred in response to declining demand. Likewise, Nixon et al. (2004) found that downsizing associated with firm reallocation strategies resulted in superior returns. In addition, the findings of other studies (Elayan et al., 1998; Hallock, 1998; Lee, 1997; Worrell et al., 1991) suggest that the negative effects of downsizing are more pronounced when downsizing involves permanent (rather than temporary) layoffs. Interestingly, Franz et al. (1998) found that downsizing by healthy firms resulted in the highest negative CARs. In sum, it appears that while downsizing is generally associated with negative wealth effects, information surrounding layoff announcements has an important influence on market perceptions.

Compared with the body of empirical literature on the wealth effects of employee downsizing, studies examining the effects of downsizing on long-term stock returns have been relatively few (Cascio & Young, 2003; Cascio, Young, & Morris). The limited evidence indicates that although downsizing has a negative impact on stock returns in the year of downsizing, the returns turn positive in subsequent years, suggesting that the benefits of downsizing in the form of improved stock returns often take some time to materialize. However, Perry and Shivdasani (2005) found no relationships between downsizing and stock returns, whereas De Meuse et al. (2004) observed a lower market to book ratio resulting from employee layoffs.

Link 3-5b: Relationships between employee downsizing and firm profitability. A number of studies in Stream IV have sought to examine whether downsizing leads to improved firm profitability. Findings, however, have been mixed. Studies by Bruton, Keels, and Shook (1996); Chen et al. (2001); Espahbodi, John, and Vasudevan (2000); Kang and Shivdasani (1997); Palmon et al. (1997); and Yu and Park (2006) suggest that employee reductions lead to performance improvements. Interestingly, Espahbodi et al. (2000); Kang and Shivdasani (1997); Palmon et al. (1997); and Perry and Shivdasani (2005) observed that such improvements occurred 2 to 3 years after downsizing, reinforcing the view that benefits from employee reductions, if any, are experienced only in the long term. However, several studies, including those by Cascio et al. (1997); Cascio and Young (2003); De Meuse, Vanderheiden, and Bergmann (1994); Suárez-González (2001); De Meuse et al. (2004); and Guthrie and Datta (2008), have found that that employee downsizing has a deleterious effect on organizational profitability.

The equivocal findings on the relationship between downsizing and profitability have motivated researchers to search for moderators in an effort to explain performance differences. Love and Nohria (2005), for example, found that benefits associated with downsizing were greater in broadly scoped and proactive downsizings and when downsizing firms were characterized by high levels of slack. In addition, Yu and Park (2006) observed that although downsizing resulted in improved return on assets (ROA) among firms that did not experience losses in the 3 years prior to downsizing, no such improvements occurred among loss-making firms. In another study, Guthrie and Datta (2008) examined the moderating role of industry conditions on the downsizing-performance relationship and concluded that the negative effects of downsizing on organizational performance were more pronounced in industries characterized by high R&D intensity, growth, and low capital intensity. Finally, in a study of downsizing in hospitals, Chadwick, Hunter, and Walston (2004) found that HR policies and practices (e.g., advance notice, extensive communication, provision of benefits) had a positive effect on (perceptions of) downsizing success and financial performance.

Link 3-5c: Relationships between employee downsizing and other organizational outcomes. In addition to examining financial implications, research belonging to Stream IV has also examined the relationships between employee reductions and sales growth, labor productivity, R&D/technological investments, and corporate reputation. Palmon et al. (1997) found that the positive impact of downsizing on sales was greater in layoffs characterized as "efficiency enhancing" as opposed to "declining demand." Wayhan and Werner (2000), however, observed no significant relationship between downsizing and sales growth in the long term. Although sales growth improved in the 4 years following downsizing, it regressed in the following years. Equivocal findings also characterize the research on the effects of downsizing on labor productivity. Although Chen et al. (2001) and Chalos and Chen (2002) observed increased productivity following downsizing, no such relationship was found by Mishra and Mishra (1994); Said, Le Louarn, and Tremblay (2007); Suárez-González (2001); and Yu and Park (2006). The findings of Wagar (1998) suggest that permanent workforce reduction results in lower organizational efficiency. In addition, Zatzick and Iverson (2006) present a study of labor productivity effects resulting from the interaction of layoffs and the use of high-involvement work practices (HIWPs). They found that firms with more extensive use of HIWPs experienced lower productivity as a consequence of downsizing. Interestingly, firms that maintained their investment in HIWPs following layoffs were able to stem productivity losses.

Questions related to the effects of employee downsizing on firm R&D have been addressed by Bruton et al. (1996), Dougherty and Bowman (1995), and Espahbodi et al. (2000). Although findings indicate that downsizing generally results in lower firm R&D expenditures and reduced organizational innovation, Bruton et al. (1996) found that such reductions did not diminish organizational performance. In a related study, Amabile and Conti (1999) found that downsizing was accompanied by negative changes in the work environment, which, in turn, contributed to reduced creativity. Finally, the findings of two studies belonging to Link 3-5c (Flanagan & O'Shaughnessy, 2005; Zyglidopoulos, 2005) on the effects of layoffs on firm reputation indicate that layoffs have a detrimental effect on firm reputation. In addition, Flanagan and O'Shaughnessy (2005) found that the negative impact

was stronger in the context of newer firms than older firms. Interestingly, whereas Flanagan and O'Shaughnessy (2005) did not find any moderating effects of prelayoff ROA, Zyglidopoulos' (2005) study found that firm profitability intensified the negative effects of layoffs on firm reputation.

Summary and Implications for Future Research

Although extant research has made significant progress in understanding issues surrounding the phenomenon of employee downsizing, much remains to be done. Based on the preceding review of empirical research spanning 25 years, very little can be said with certainty regarding the antecedents and consequences of employee downsizing. At the same time, however, several implications for future research can be identified. These are discussed in the following paragraphs.

Implications for Theory Building

As noted earlier, a plethora of theoretical frameworks have been invoked to study downsizing. This theoretical potpourri is perhaps inevitable given the variety of academic disciplines and research questions represented in this literature. Depending on their discipline-informed perspective and level of focus, scholars have invoked individual-, group-, and organizational-level theories. However, the modal downsizing study invokes a single overarching theoretical perspective. Future studies using multitheoretic perspectives and addressing downsizing at multiple levels would greatly enrich knowledge. Moreover, studies of downsizing consequences, for example, tend to focus on either individual or organizational outcomes. Studies that focus on organizational-level outcomes are often "black box" studies that assume and argue that downsizing affects organizations through cumulative effects on individual employees but do not measure these individual-level intermediaries. For example, Guthrie and Datta (2008) argued that industry conditions will moderate the impact of downsizing on firm performance because of differential effects on employees' social capital, psychological contract perceptions, and so on, but did not measure these employee-level outcomes. Similarly, Chadwick et al. (2004) argue that HR practices (e.g., advance notice, extensive communication) used by downsizing firms can affect firm performance through their effects on employees' commitment levels and organizational citizenship behavior but do not measure these latter individual-level constructs. Downsizing studies at the individual level find that certain HR practices (e.g., severance pay, insurance coverage) increase perceptions of organizational trust and support but do not examine the implications of these employee outcomes on organizational functioning and performance. Future studies of downsizing that combine both individual- and organizational-level outcomes would provide a much richer and nuanced understanding of downsizing effects. Such studies might usefully employ hierarchical linear modeling to accommodate multilevel measures and designs.

Inconsistencies in research findings can be partly attributed to the use of underspecified models, raising obvious questions regarding validity and generalizability. For example, as

indicated by some of the moderating variables identified in our review, negative outcomes associated with downsizing may not solely reflect a problem with downsizing per se but instead be reflective of conditions under which downsizing was initiated or the manner in which it was implemented. Thus, negative outcomes may at least partly result from a failure on the part of the firm to meet the organizational requirements of effective employee downsizing, suggesting the need to delve into more complex relationships in the examination of downsizing outcomes. Given the critical role of environmental and organizational factors, the omission of such variables in research models can significantly reduce the contribution of studies in furthering our understanding of the true consequences of downsizing.

Similar to the either–or approach with regard to individual versus organizational outcomes, our review indicates that relatively few studies incorporate both antecedents and consequences in a single theoretical framework. Future studies should develop theoretical frameworks that seek to understand consequences in the context of antecedents. The effect of downsizing on individuals or organizations may be conditioned by precipitating factor(s). For example, although the strategic management and the industrial organization literature considers a firm's industry to have an important impact on both the incidence and effectiveness of organizational initiatives, the role of the industry has been underrepresented in HR research. This is certainly true in the context of employee downsizing. Yet as the recent study by Guthrie and Datta (2008) highlights, the industry context plays an important role. Future studies examining both the antecedents and consequences of downsizing in a single design can help integrate and bridge earlier work such as this.

Implications for Research Methods

Our review of extant research suggests that downsizing research suffers from a number of methodological limitations. Perhaps most important, there is limited consistency in the conceptualization and measurement of key constructs. Of particular concern is the ambiguous operationalization of the construct of employee downsizing. Studies have used a variety of approaches, including (a) newspaper announcements, (b) surveys that allow firms to self-identify themselves as "downsizers" based on various criteria, and (c) the use of publicly available data (e.g., Compustat) to identify "downsizers" based on (various) levels of employment level change (e.g., 3% decrease, 5% decrease). Many organizational-level studies apply some type of decision rule to employment-level change data to dichotomize firms as either "downsizers" or "nondownsizers," with the decision rules varying across studies. In contrast, others have used percentage change in the number of employees as a direct and continuous measure. Still others (e.g., Iverson & Zatzick, 2007; Mishra & Mishra, 1994) have used Likert-type scales in the context of survey-based data collection. The diversity of approaches is somewhat troubling and problematic. It is not clear that these various operationalizations are equivalent (i.e., measuring the same underlying construct), making comparisons across studies difficult. We believe that the methodological heterogeneity that characterizes downsizing research (coupled with conceptual differences) continues to constrain the advancement of this literature. Indeed, the downsizing literature would benefit significantly from a more careful consideration of the construct validity of "downsizing." Future work providing psychometric evidence (e.g., reliability, convergent and divergent

validity) of downsizing measures would increase confidence in both the internal and external validity of these studies. Included here would be a consideration of the parameters and dimensionality of the downsizing construct, incorporating issues such as speed, scope, frequency, and so on.

Our review of the empirical literature also reveals that a vast majority of studies have limited themselves to the examination of direct effects. Only a few have examined the moderating effects of contextual factors in understanding determinants and outcomes of employee downsizing. As noted earlier, this fact may explain the equivocal findings associated with different links in the framework. For example, it can be reasonably argued that relationships between organizational antecedents and downsizing are contingent on industry characteristics (e.g., industry growth or capital intensity). In addition, future research needs to make more extensive use of mediating models wherein downsizing is seen as affecting outcomes through their effects on key implementation factors (e.g., human capital, structural characteristics, organizational culture). Greater use of moderating- and intervening-effects models should provide interesting insights and help improve our understanding of the downsizing phenomenon.

An important limitation of extant research is the overreliance on static, cross-sectional designs. These designs limit the ability to infer causality, especially the ability to tease out the complex recursive interrelationship between constructs treated as both antecedents and consequences (e.g., firm performance) over the course of time. Organizational- and individual-level effects of downsizing may vary substantially among infrequent versus recidivist downsizers. If downsizing in a particular firm negatively affects individuals and firms, is this effect diminished or magnified over the course of multiple downsizing events? Although preliminary evidence exists (Gilson et al., 2004), further inquiry is necessary. Consistent with the views of Cappelli (1999), frequent downsizing events may alter the parameters of the psychological contract so that downsizing events within a particular firm, or within the context of multiple downsizings occurring in the relevant external marketplace, may prove less deleterious across time. With respect to both individual- and firm-level outcomes, is the effect of multiple smaller events cumulatively equivalent to less frequent, but larger events? Are these effects modified by antecedent conditions (e.g., level of firm performance, industry conditions)?

Another interesting issue is the possibility of nonlinear effects. Some of the equivocal findings revealed in this review may result from an overreliance on linear models and linear estimation techniques. Especially, in studies of downsizing consequences, it seems likely that curvilinear relationships may exist. At the organizational level, one possibility is an inverted-U relationship, with low levels of downsizing (under the "right" conditions) having positive effects, whereas higher levels of downsizing might remove "muscle" as well as "fat," leading to diminished or negative returns. Examining for nonlinearity in downsizing in terms of both magnitude and frequency across time would allow for greater precision in addressing questions such as these. Researchers might fruitfully employ real-time longitudinal designs, wherein researchers record the timing and other characteristics of downsizing events and activities as they occur within the organization through regularly scheduled and intermittent real-time observations. Methodologies such as event history analysis might also generate insights into the longitudinal effects of the timing and spacing of downsizing over a period of time.

Along with hierarchical linear modeling, our suggestion that future research designs should incorporate multiple measures (e.g., antecedents *and* consequences; individual- *and* organizational-level outcomes) indicates the need for greater use of structural equation modeling. Use of structural equation modeling, for example, would allow downsizing and various downsizing characteristics to be included as both endogenous and exogenous variables. As endogenous variables, downsizing events and attributes could be predicted or determined by environmental or organizational factors. However, they would also serve as exogenous variables, affecting both individual and organizational outcomes. These relationships can then be assessed through a system of simultaneous, structural models. Understanding downsizing as a multilevel, complex phenomenon will also require mixed use of both quantitative and qualitative methodologies.

In addition to the United States, extant research has explored downsizing in a variety of national settings, including, among others, Japan (Ahmadjian & Robbins, 2005), Russia, Ukraine, and Belarus (Filatotchev et al., 2000), the United Kingdom (Conyon et al., 2002; Yawson, 2006), China (e.g., Frazier, 2005), Australia (e.g., Iverson & Zatzick, 2007), Spain (Suárez-González, 2001), Finland (Lehto & Böckerman, 2008), and Belgium (Coucke et al., 2007). Although these studies provide insights into the generalizability of findings across national borders, future research should attempt to include international samples in their research designs to directly compare downsizing antecedents and consequences associated with institutional and cultural differences. These designs will be particularly useful in examining and comparing longitudinal changes in the context of a world economy that is both globalized and dynamic. Similar to studies of other HR policies and practices (e.g., Pudelko & Harzing, 2007), these designs will contribute to the convergence versus divergence debate. Moreover, research designs that allow for the examination of cross-national comparisons of downsizing within the confines of multinational organizations can contribute descriptively and normatively to the standardization versus localization debate found in the international management literature.

Future Research Topics

In addition to the need for research invoking alternative theoretical perspectives and designs, there are a number of downsizing topics in need of future attention. These topics include both a revisitation of previous research questions and a consideration of research questions not previously addressed. This section provides guidance with respect to both these types of questions.

Although journal editors are generally less receptive to studies on research questions that have been examined previously, our frequent use of the term *equivocal* in describing extant findings suggests that a number of questions remain unresolved. In terms of environmental antecedents, although slackening industry demand, privatization, and deregulation seem to promote downsizing, findings are more disparate with regard to the role of other industry characteristics. As such, future work should carefully address characteristics such as capital/labor intensity, industry competition, market power, and changes in the technological

environment. In addition, most downsizing studies have used samples drawn from the manufacturing sector. More recently, however, there have been very high levels of downsizing in the service sector. Are the factors that influence downsizing in the service sector any different from those in the manufacturing sector? Do the organizational implications from a performance standpoint differ? Are the downsizing effects on social networks more deleterious in "high-touch" and/or "high-tech" service environments?

In terms of organizational antecedents, those that have been studied but require further analysis include firm size, age, business strategy, and ownership/governance structures. However, studies are also equivocal with regard to findings on the relationship between employee compensation level and downsizing. One proviso here is that the relevant construct to study is compensation *costs*, as opposed to compensation *rates*. Although a given firm may have higher per-capita wage rates, their compensation expenses may, in fact, be lower—depending on the productivity levels of their workforce. This may partially explain previous equivocal findings. In terms of the role of competitive strategy, Coucke et al.'s (2007) finding linking differentiation strategies with lower levels of downsizing is consistent with previous work showing that "differentiators" make greater investments in employees via high-performance work systems (Guthrie, Spell, & Nyamori, 2001). As in this latter work, future downsizing research should investigate whether or not competitive strategy interacts with downsizing to predict individual or organizational outcomes. In terms of top management teams (TMTs) and governance, future work might fruitfully investigate the resources that owners, directors, and executives bring to the firm that might shape the timing and character of downsizing events. What role—if any—does the board play in monitoring, rewarding, and/or disciplining CEOs and executives for effective or ineffective downsizing decisions? Similarly, what role—if any—do institutional investors play? In addition, we encourage continued investigation of the role of foreign investors, a powerful group whose views on downsizing may conflict with a firm's management team's views. Moreover, arguments advanced in the upper-echelons literature suggest that although the CEO undoubtedly plays a leading role, organizational decision making is rarely the exclusive domain of a single individual. As such, studies should more broadly assess the role of executive teams in downsizing. As part of this research, a fertile area might be the relationship between top management team characteristics and the manner in which downsizing is conducted. For example, TMTs dominated by executives with finance or accounting backgrounds may be less inclined to emphasize or attend to process issues during layoffs.

Turning next to topics related to consequences, as alluded to earlier, it is useful to think of downsizing in the broader context of organizational-change initiatives. It is axiomatic that change is hard, and this is especially true of change involving reductions in jobs and people. As our earlier review of the research on outcomes indicates, like many change efforts, downsizing often fails to live up to expectations. From an organizational change perspective, both "process" and "contextual" factors likely influence the extent to which downsizing yields positive or negative outcomes (Self et al., 2007). A number of issues on downsizing consequences are in need of further research.

Research is needed on the relationship between involuntary turnover through layoffs and subsequent or concurrent voluntary turnover. Downsizing is typically undertaken in an

attempt to improve organizational performance, yet it seems to promote increased voluntary turnover (Trevor & Nyberg, 2008). Because research has confirmed that voluntary turnover consistently harms organizational performance (Shaw, Gupta, & Delery, 2005), gaining further insights into this phenomenon would seem an important area for additional research. In particular, research identifying the performance categories of those electing to leave would be illuminating. Are the highest performers more likely to leave because they have the best labor market mobility? Alternatively, perhaps lower performers—seeing themselves as future targets for being downsized—are more likely to depart. Trevor and Nyberg (2008) identified job individual (job embeddedness) and organizational characteristics (HR policies) that influence turnover in the context of downsizing. Of interest would be the identification of additional factors that also moderate the propensity for survivors to "jump ship." Examples might include individual characteristics such as personality as well as internal organizational factors such as relative pay rates of the employer and communication processes. External factors such as the extent of downsizing by other employers in the relevant industry environment may also play a role. Regarding communication, although the importance of effective communication during downsizing has received some attention, more research seems warranted.

Cappelli (1999) maintains that because job insecurity has become so commonplace, its negative effect on employee attitudes and behavior has reduced substantially. The widespread use of layoffs in the latter part of 2008 and into 2009 provides a forum to test these notions. Even in the midst of widespread layoffs, however, there will be firms that engage in "countercyclical" staffing practices during economic downturns (Greer & Ireland, 1992). How do contrarian practices—maintaining or even increasing employment when other firms are downsizing—affect employee attitudes and behaviors and organizational outcomes? Given the global reach of the recession and downsizing, another research question would be, "To what extent do institutional and other cultural differences moderate both downsizing frequency and effects at multiple levels?" For example, how much do factors such as individualism–collectivism and uncertainty avoidance affect the use and impact of downsizing?

Future work on organizational outcomes should focus on the notion of "strategic" and "nonstrategic" employee groups. The work of both Lepak and Snell (1999) and Delery and Shaw (2001) suggest that the organizational impact of employee retention will vary depending on whether employees are part of a firm's "strategic core." According to this perspective, firm performance may be more or less affected depending on whether or not "strategic" employees are "made redundant." Based on these theoretical frameworks, future studies might identify the "strategic" and "nonstrategic" employee groups within firms or industries and study the firm effects of unstable employment relationships across these groups.

In the recent economic downturn, in addition to downsizing, firms are also turning to a range of options, including reduced work hours, pay cuts, furloughs, benefit reductions, and so on. Researchers should avail themselves of the current expansive "natural laboratory" to directly examine and compare the implications of these options vis-à-vis downsizing for both individual and organizational outcomes. How do these alternative options affect employee attitudes and behavior? Although the costs of alternative options may be higher, do accrued benefits outweigh additional costs? An additional issue for consideration is the impact of downsizing on intermediate outcomes such as compensation costs, customer

service, and product quality. Pfeffer (1998), for example, has argued that firms that "follow the herd" in terms of employment boom-and-bust cycles will fall into the trap of "buying high and selling low" in the labor market, yielding higher compensation costs relative to those firms following more stable employment patterns. Reichheld (1996) is also an advocate of enhanced employment stability, arguing that "loyalty" to employees will often spill over to customers, yielding a more loyal customer base (the so-called loyalty effect). These interesting arguments can and should be invoked in future studies of downsizing effects.

Conclusion

Restructuring, rightsizing, streamlining, making redundant, retrenching, reduction in force, selective shrinkage, dehiring, delayering, reengineering, rebalancing, outplacing—these are just a few of the phrases or euphemisms used to describe downsizing, a phenomenon that has become a ubiquitous fact of modern-day organizational life. Some management issues appear to receive a disproportionate amount of academic research relative to the magnitude, importance, or prevalence of the phenomena in the "real world." This is clearly not the case with downsizing research. Although research on downsizing is growing, it is still dwarfed by the magnitude of this phenomenon in the marketplace. There are a host of theoretically and practically important issues in need of additional research. Now, more than ever, organizational researchers need to apply relevant methodologies to these important and complex research questions. We hope that our review informs and stimulates further work in this regard.

Notes

1. These figures, however, significantly underreport the actual number of employees losing jobs because the BLS only records layoffs involving 50 or more employees.

2. Other labels have been used to describe employee downsizing. These include "employee layoffs," "employee reductions," "employee involuntary turnover," and "job cuts." Because they tend to refer to the same phenomenon, studies that have used them are included in our review.

3. Although we attempted to be both exhaustive and inclusive in our search, it is certainly possible that we overlooked and failed to include relevant research in our review. To the extent that this is true, we apologize to the authors of any omitted papers and to the readers of this review.

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